

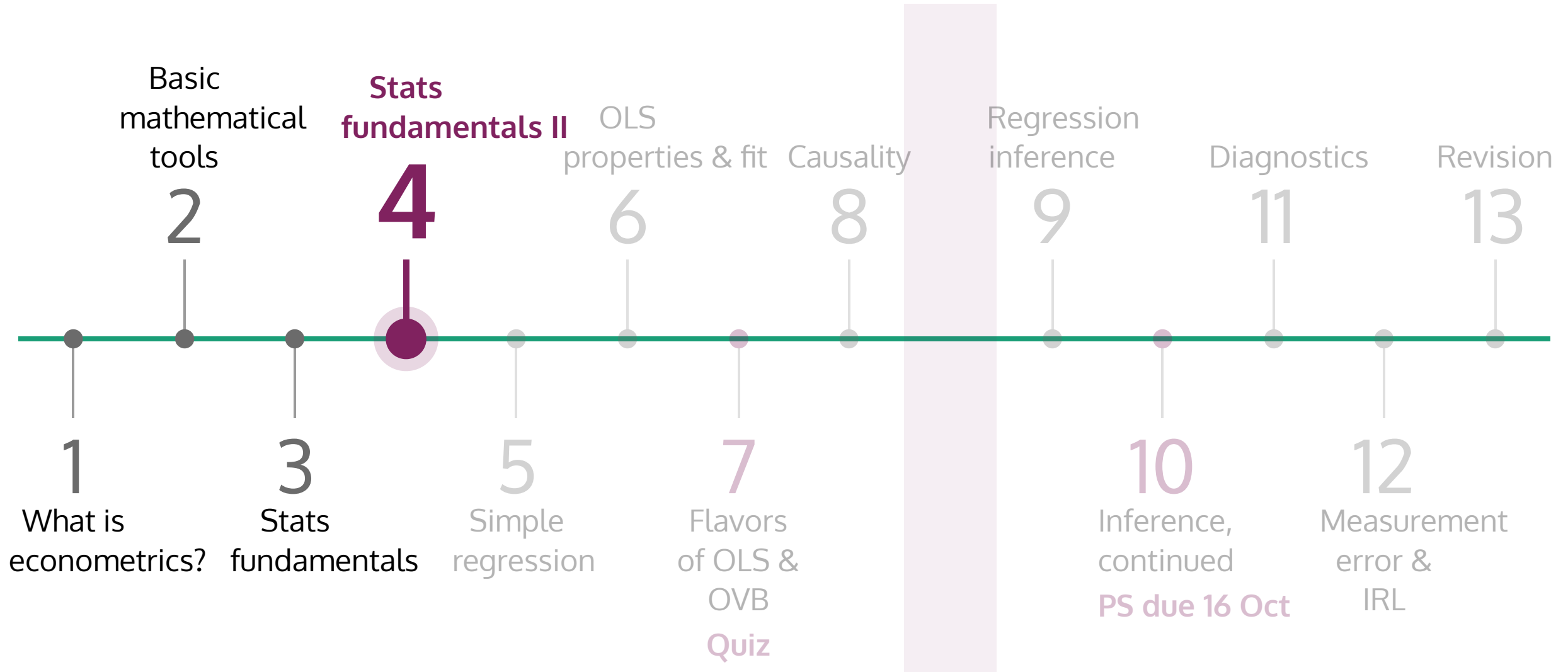


ECON 2041

Introductory Econometrics

Great expectations

Plan for the semester



Plan for today

1. Your grade forecasts, revealed
2. Income and life expectancy in the US
3. Discrete, continuous, or categorical?
4. The correlation gallery
5. Where a covariance comes from

Your grade forecasts

What I asked you

Your own semester

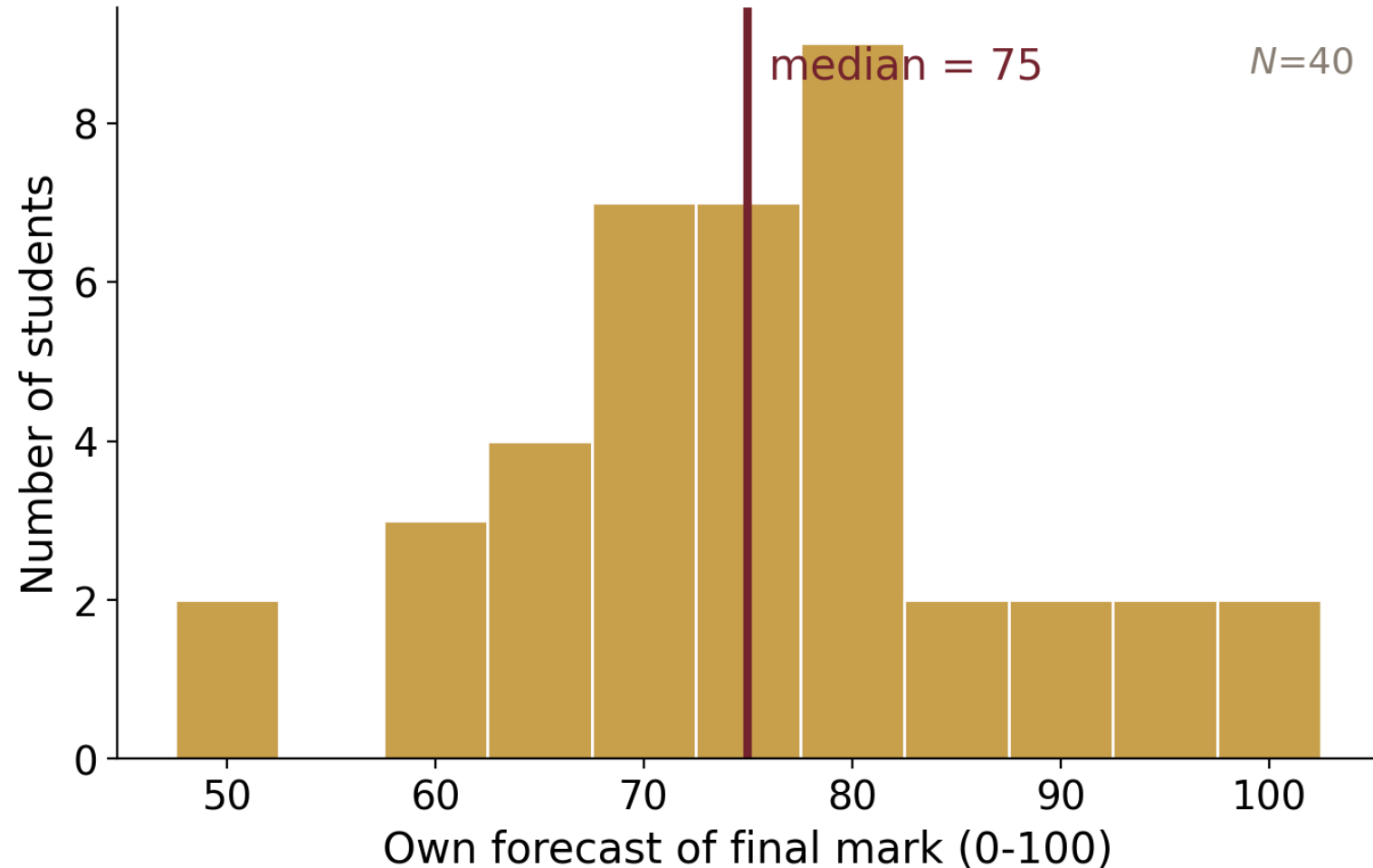
- Expected final mark (0-100)
- Plans: attend live? study hours? tutorials before class?
- The mark if you did everything

Three hypothetical students

- Attends live: rarely / some weeks / most weeks
- Study hours: under 2 / 2 to 5 / more than 5
- Tutorials first: no / yes

For each: a mark, a pass chance (50+), a distinction chance (75+)

Nobody expects to fail, and the middle of the room expects 75

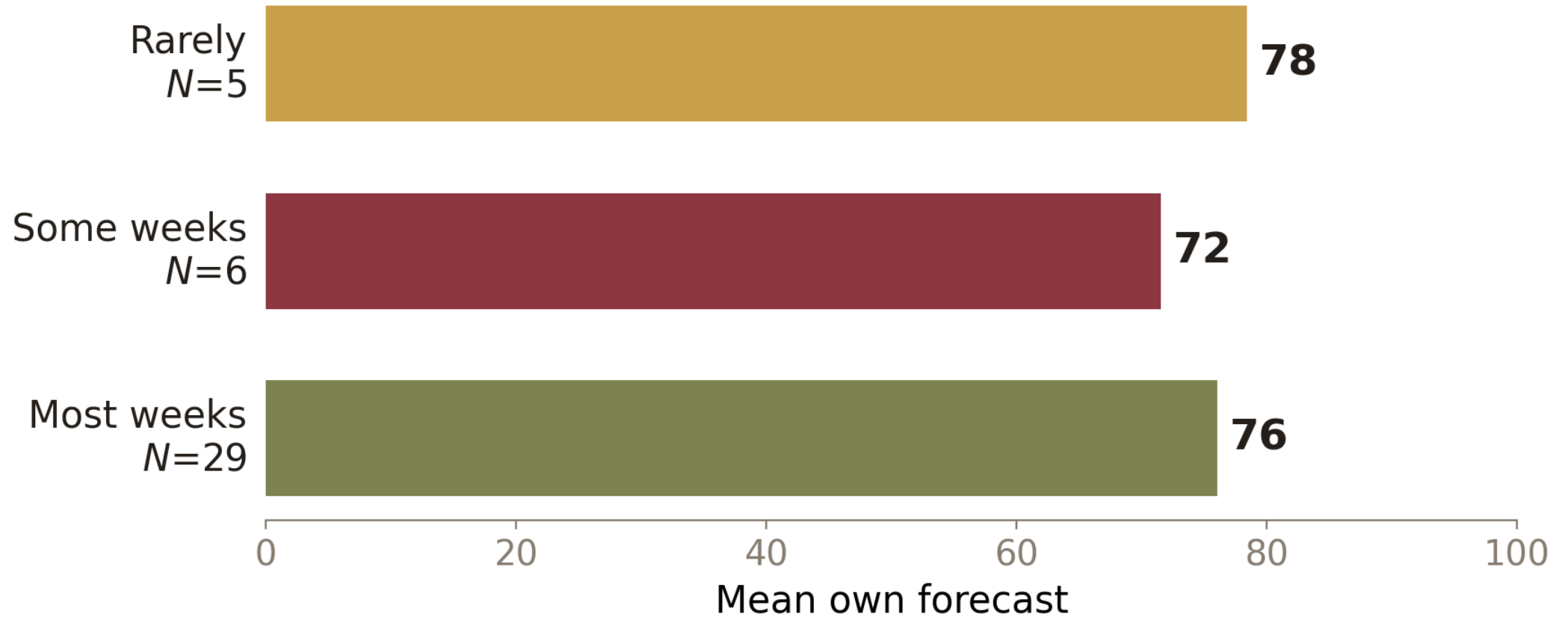


Last year's class, for comparison

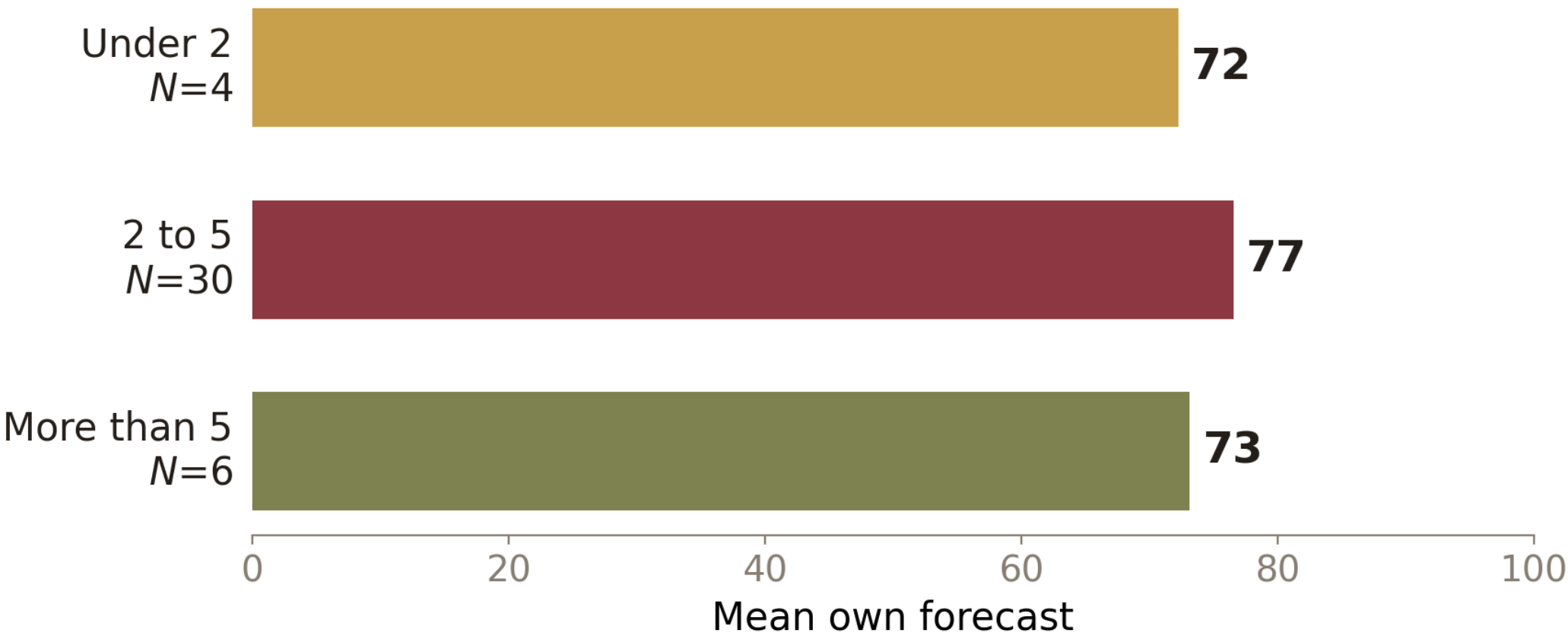
[Placeholder: distribution of final marks, ECON2041 2025]

Figure lands once the marks spreadsheet arrives

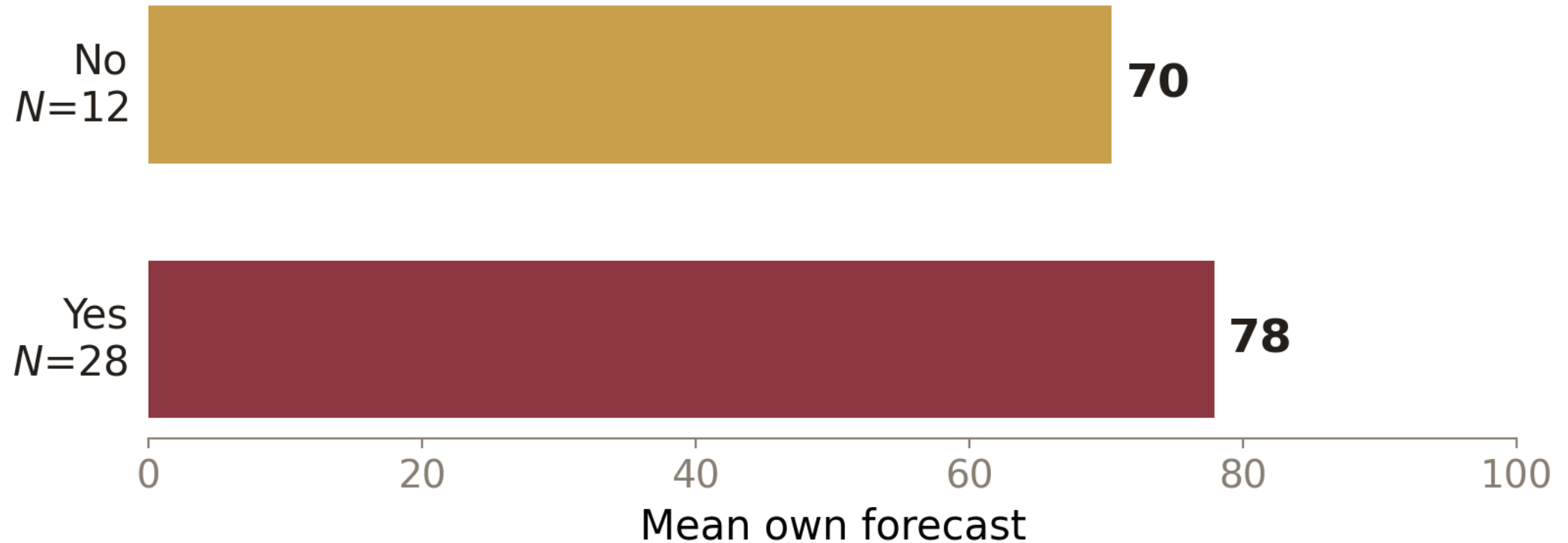
Attendance plans: forecasts flat, and two of the bars stand on five people



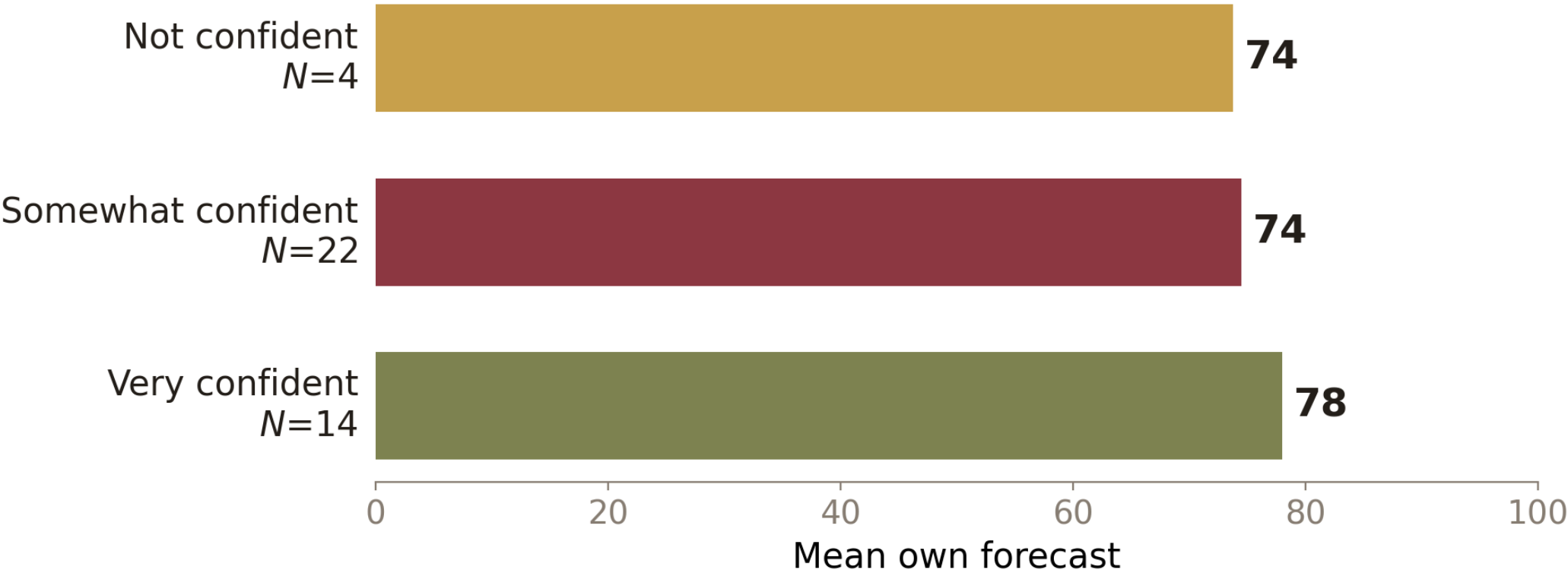
Study-hour plans: 72, 77, 73



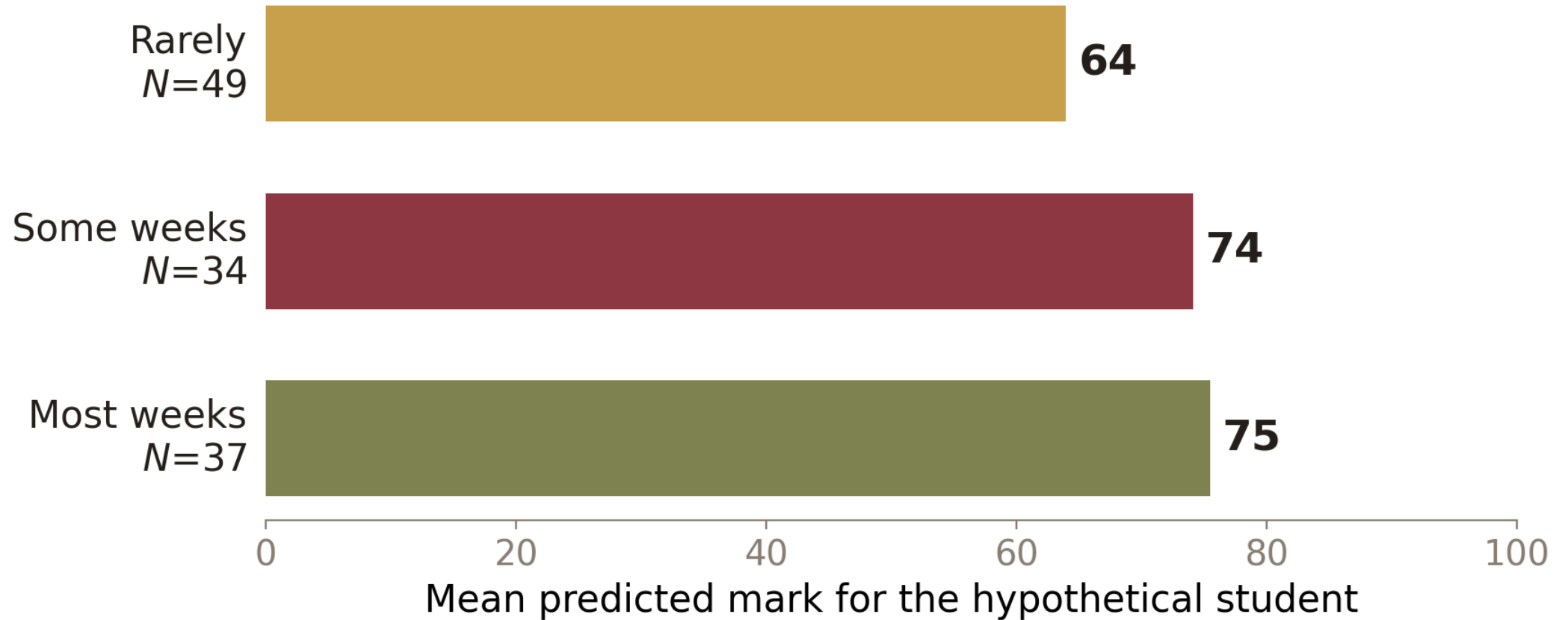
Tutorials before class: the one plan that moves the forecast



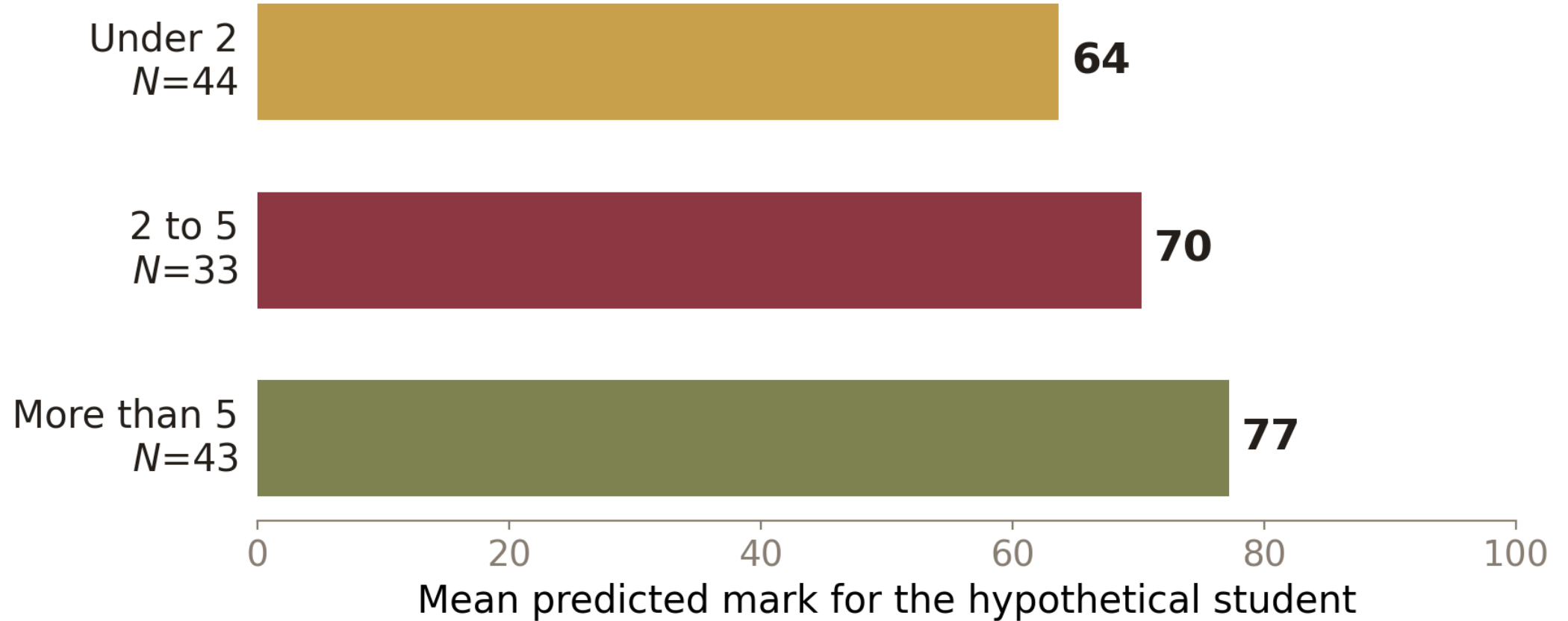
Confidence in the plan barely matters



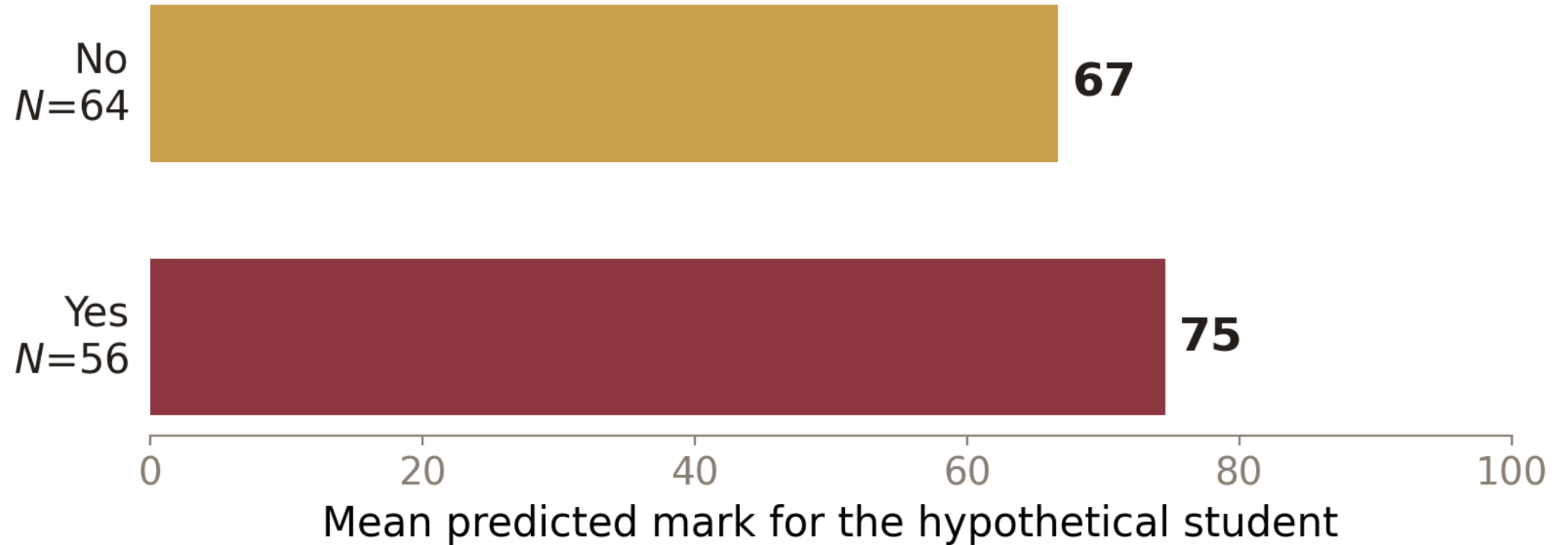
For a hypothetical student, attendance is worth 11 marks



Study hours: 64, 70, 77



Tutorials before class: 67 against 75

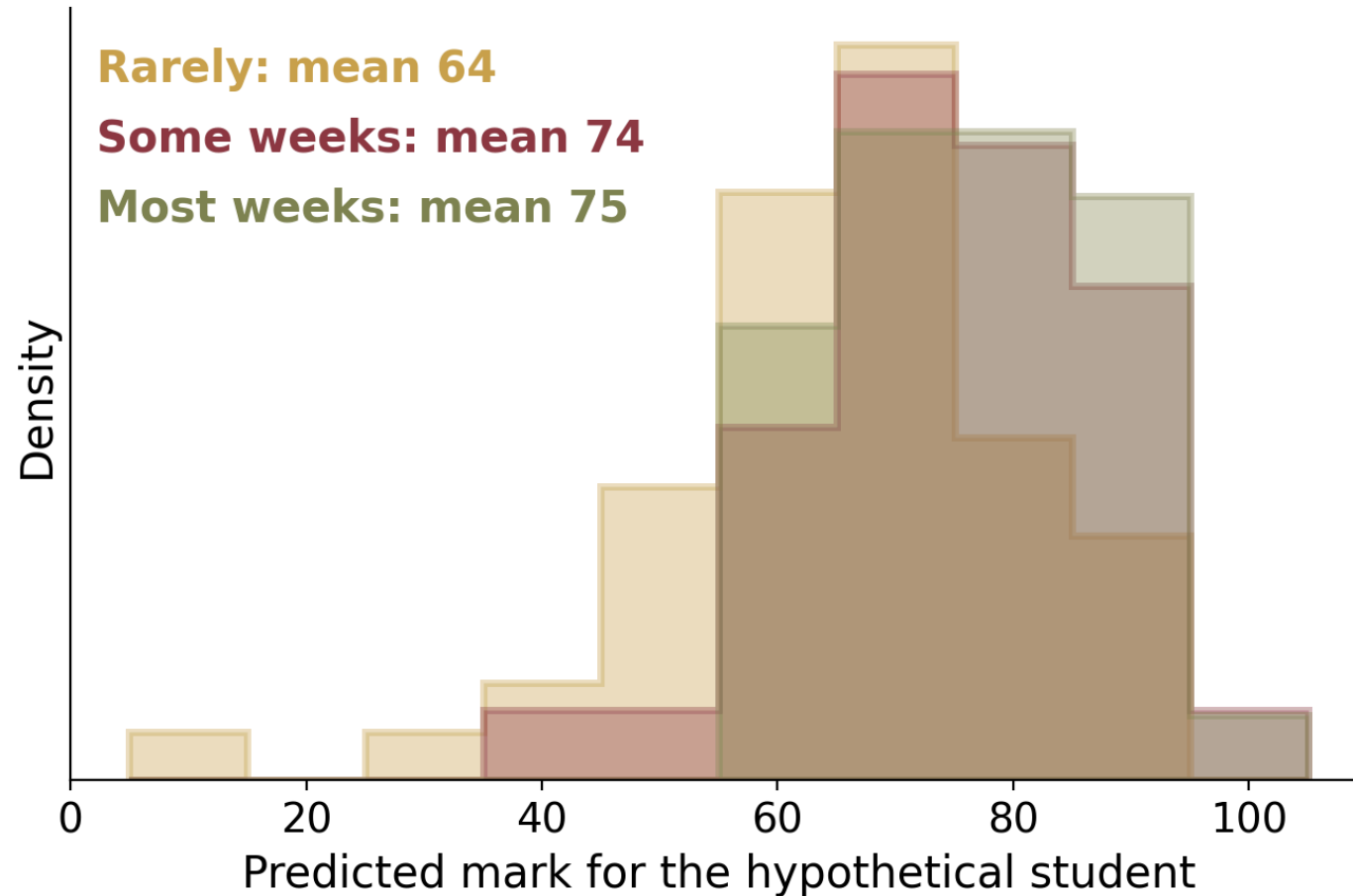


Independence by design: the portal drew each attribute on its own

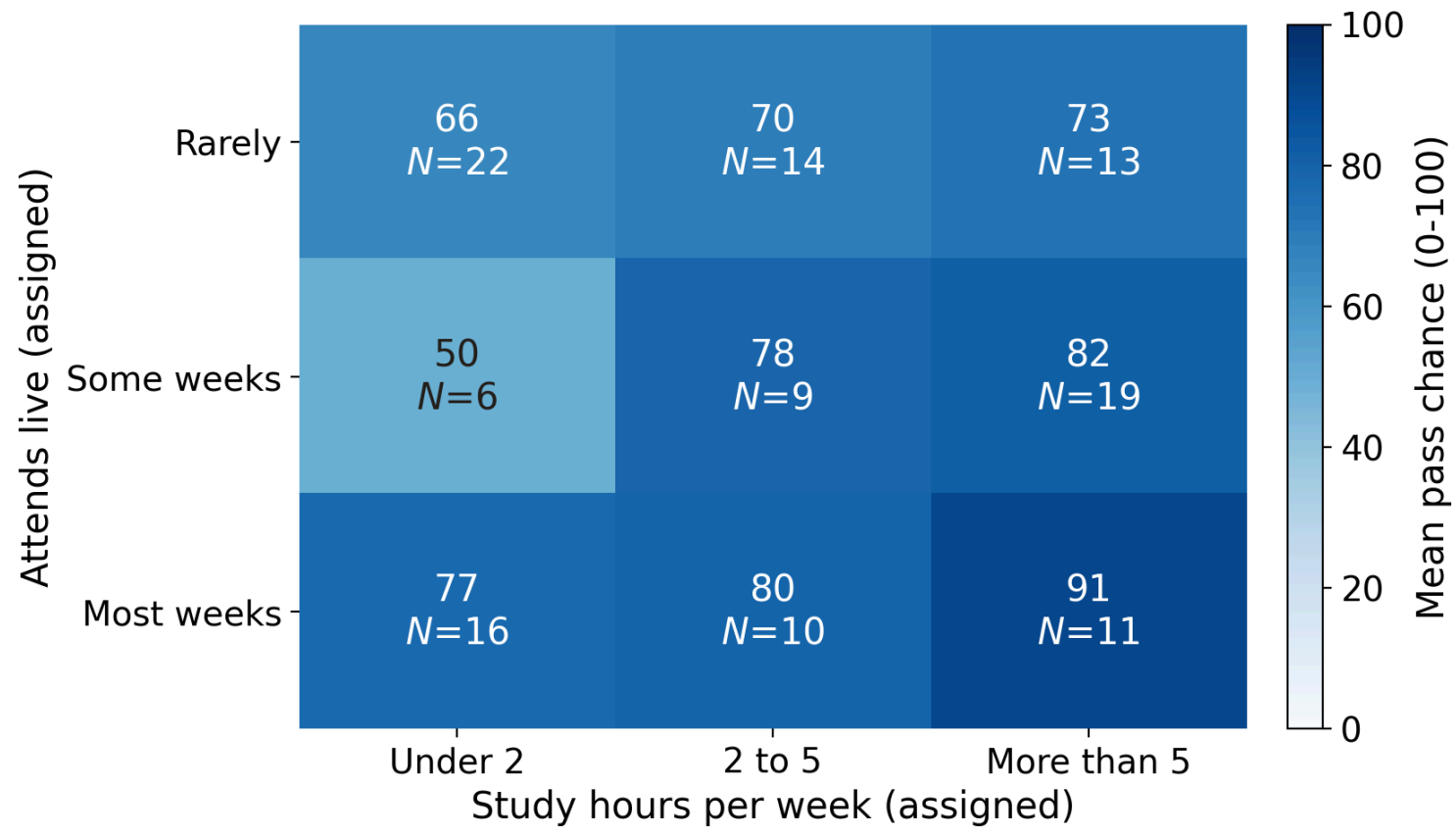
Share of hypothetical students at each study-hours level, within each attendance level

Attends live	Under 2	2 to 5	More than 5	<i>N</i>
Rarely	45%	29%	27%	49
Some weeks	18%	26%	56%	34
Most weeks	43%	27%	30%	37
All 120	37%	28%	36%	120

Predicted marks are not independent of assigned attendance



Chance of passing, given attendance and study hours



A distinction is a pass: 10 answers said otherwise

For each hypothetical student, two chances:

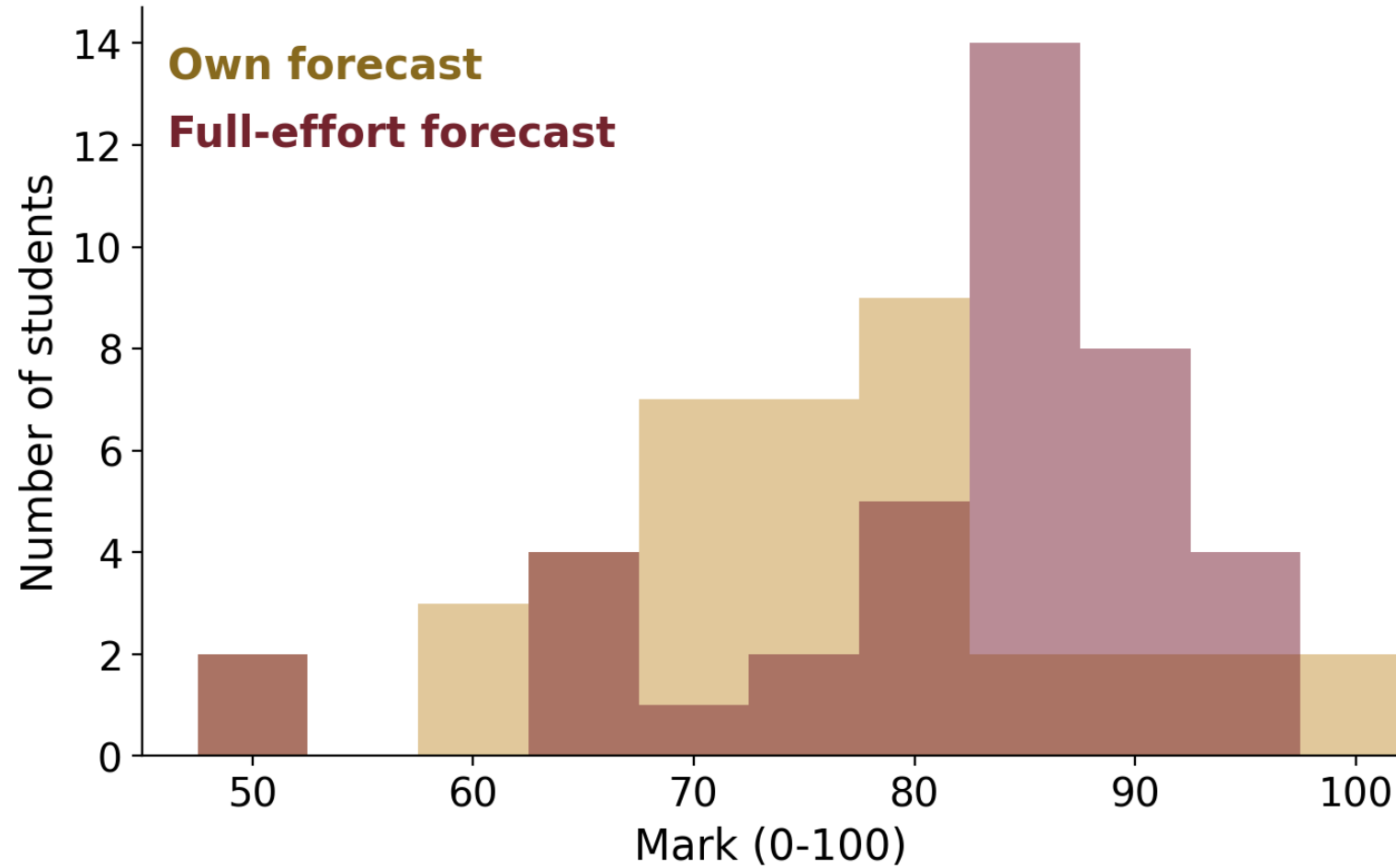
- $P(\text{mark} \geq 50)$, the chance of a pass
- $P(\text{mark} \geq 75)$, the chance of a distinction

Every mark of 75 or more is also a mark of 50 or more

$\{\text{mark} \geq 75\} \subset \{\text{mark} \geq 50\}$, so $P(\text{mark} \geq 75) \leq P(\text{mark} \geq 50)$

10 of 120 answers had the distinction chance above the pass chance

You think full effort is worth about 6 marks



Income and life expectancy

Exploratory data analysis: why?

We always start by looking at the data

- Check your understanding of the variables
- Guide subsequent analysis
- Give context to results
- Generate additional questions
- Offer simple, but possibly important answers!

Descriptives can be incredibly powerful

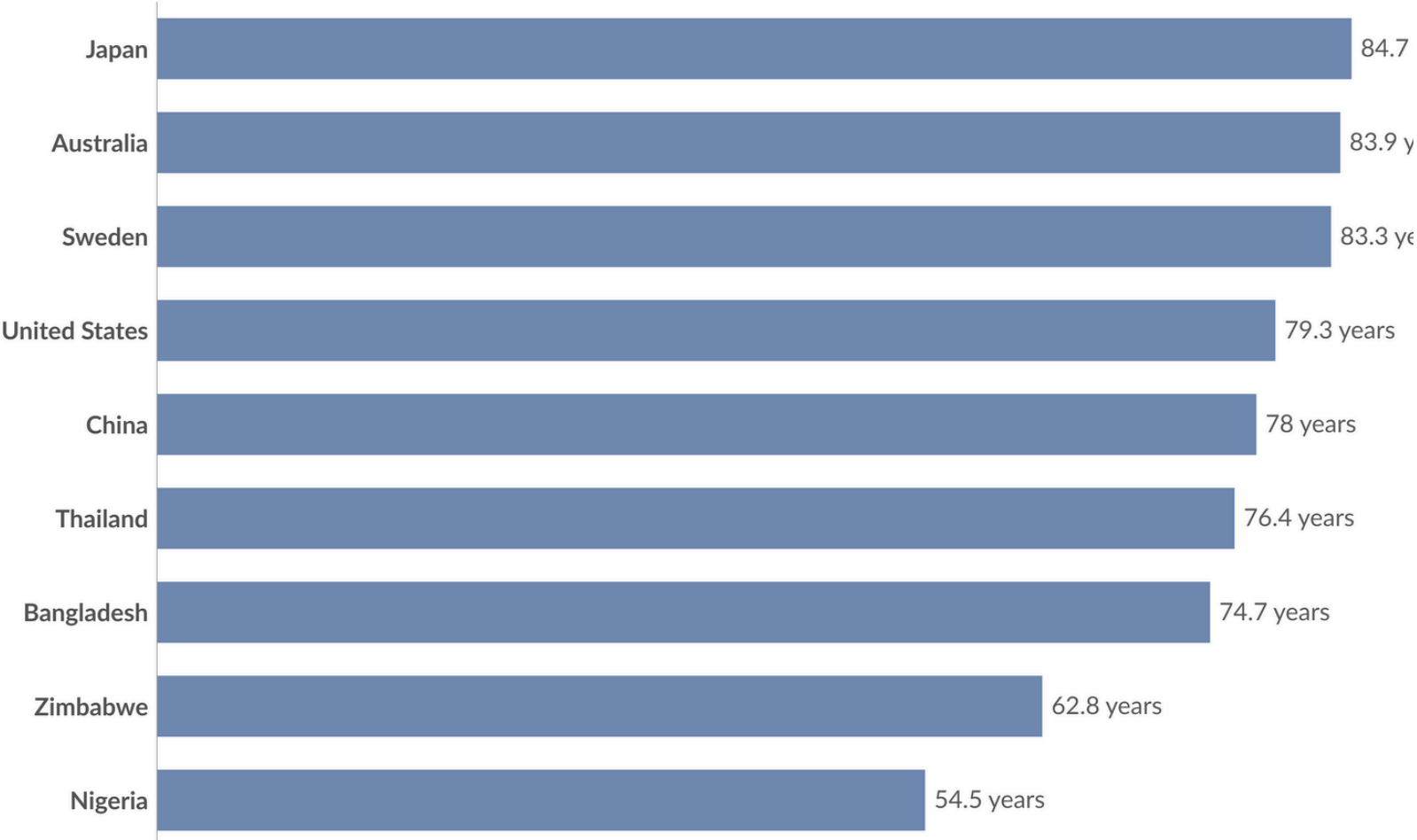
Descriptive statistics

- mean
- median
- variance
- correlations
- conditional expectations

Life expectancy at birth

Life expectancy, 2023

The period life expectancy¹ at birth, in a given year.



Life expectancy is based on an estimate of the average age that members of a particular population group will be when they die

Source: [Our World in Data](#)

Life expectancy by current age

Life expectancy for people of different ages, Australia, 2023



The total period life expectancy¹ for people who have reached a given age.



Source: [Our World in Data](#)

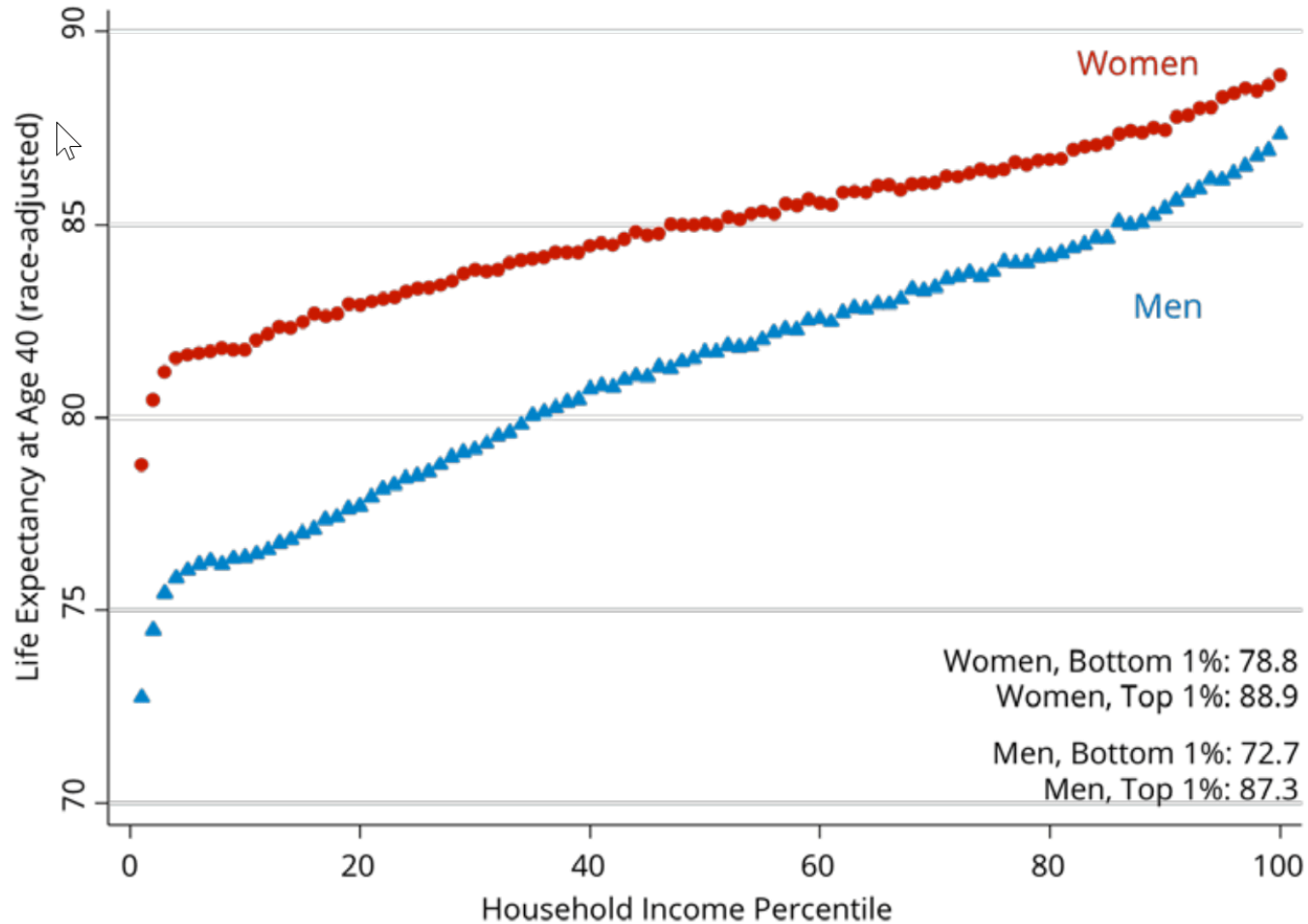
Guess the gap

US men, aged 40, 2001 to 2014

How many more years does a man in the richest 1% expect to live than a man in the poorest 1%?

Portal round open now: tell your neighbor your number first, then slide

The gap is 14.6 years



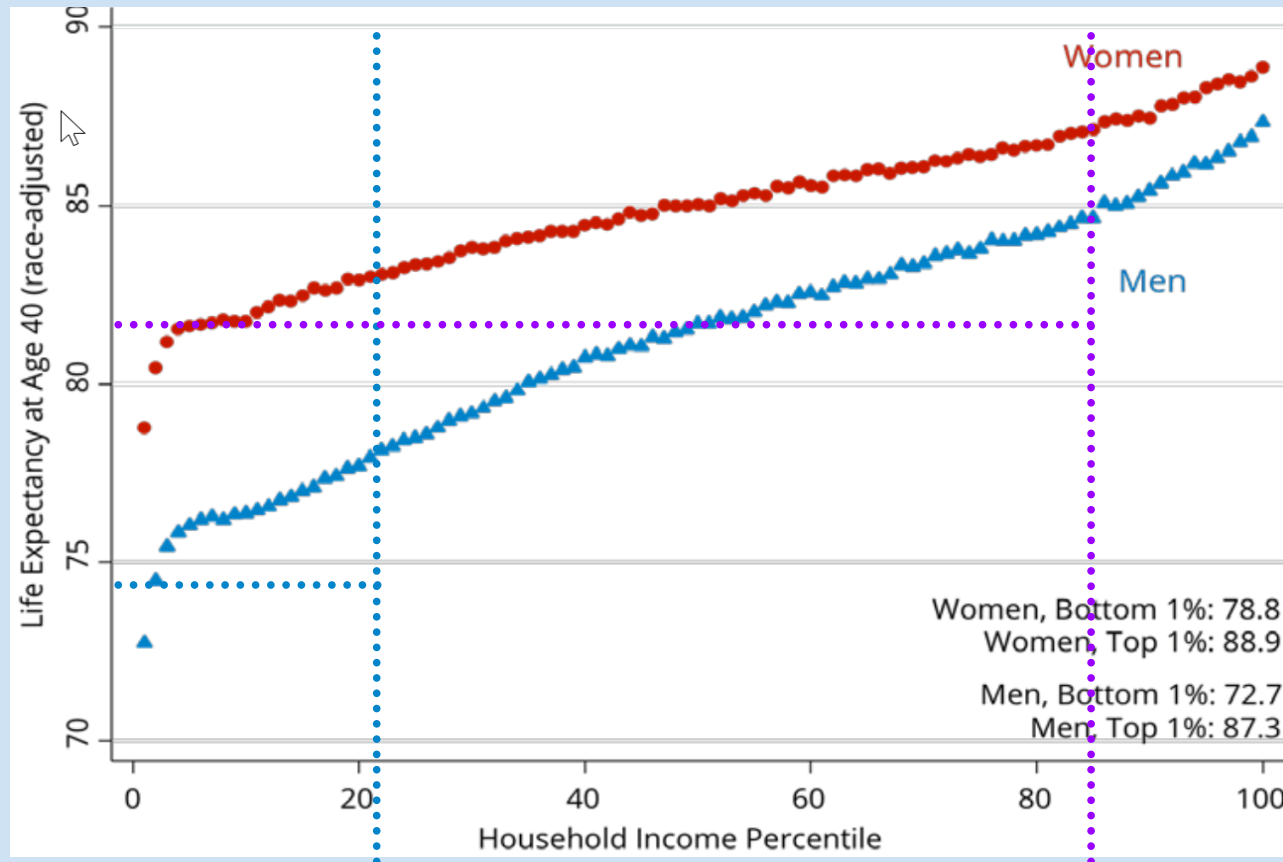
Men, age 40

- Poorest 1%: 72.7 years
- Richest 1%: 87.3 years

Women: a gap of 10.1 years

Source: [Chetty et al. \(2016\)](#)

Life expectancy in the United States



Y: life expectancy at age 40

X: income percentile

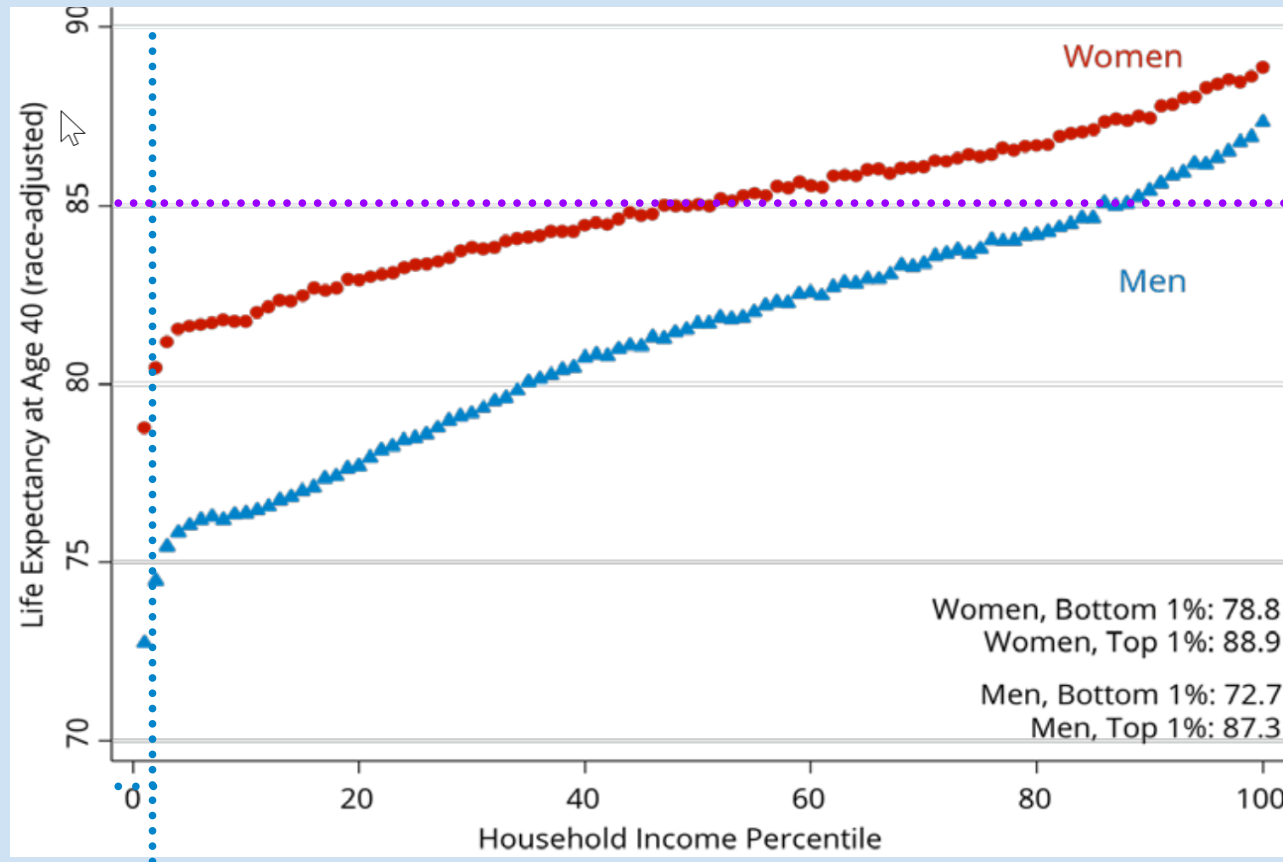
$$E[Y | X = x]$$

Compare:

- $E[Y | X = p20]$
- $E[Y | X = p80]$

Source: [Chetty et al. \(2016\)](#)

Life expectancy in the United States



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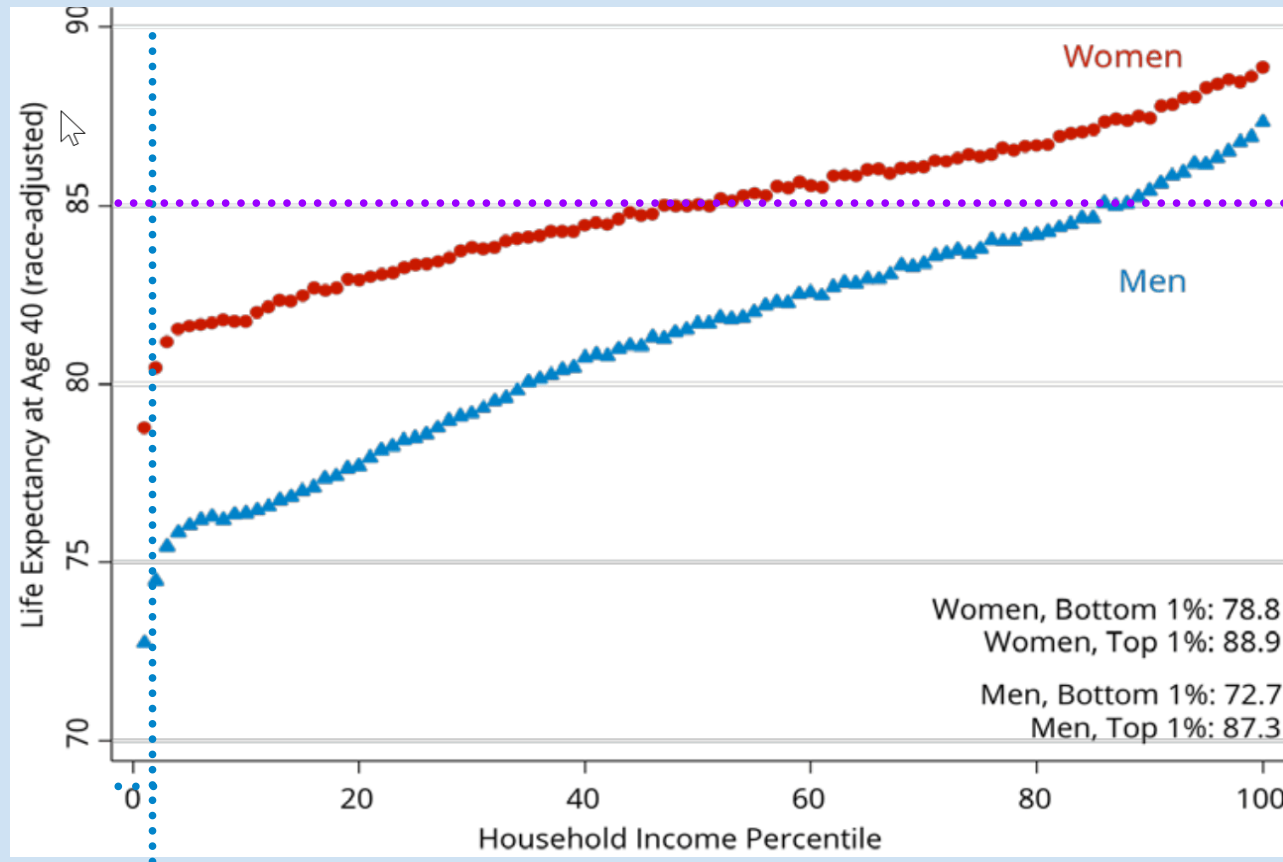
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Compare:

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Source: [Chetty et al. \(2016\)](#)

Life expectancy in the United States



Y: life expectancy at age 40

X: income percentile

$$E[Y | X = x, Z = z]$$

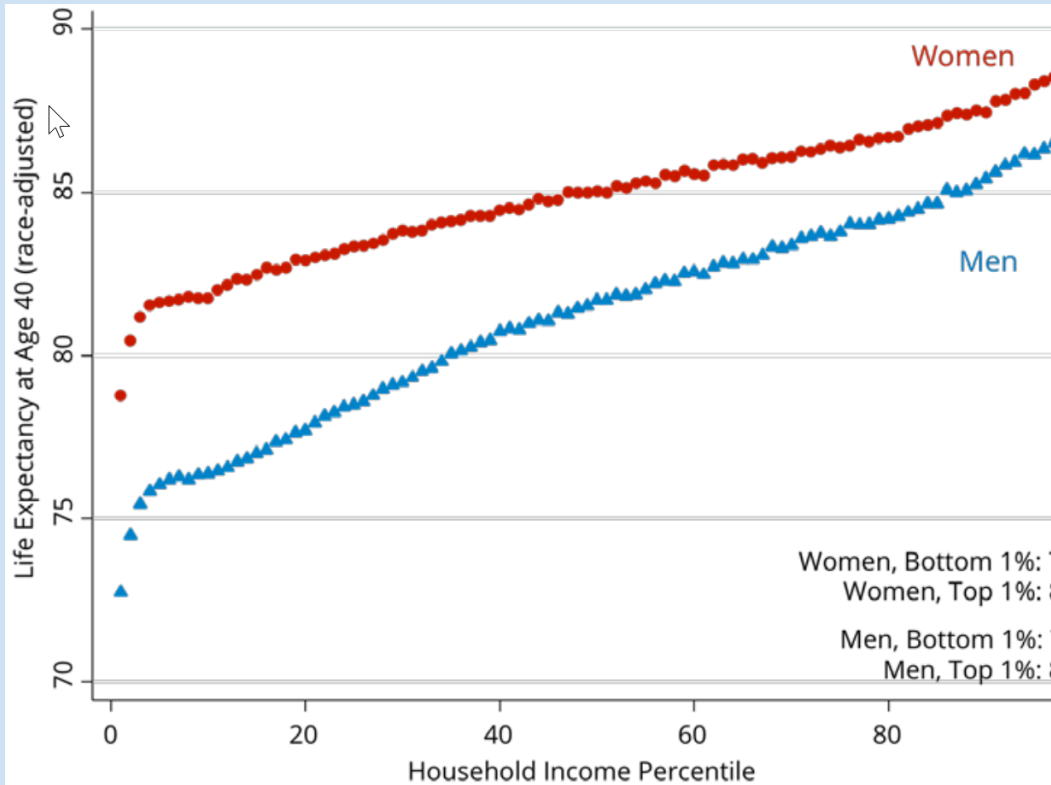
Compare:

- $E[Y | X = p20]$
- $E[Y | X = p80]$

Source: [Chetty et al. \(2016\)](#)

Unconditional vs. conditional expectation

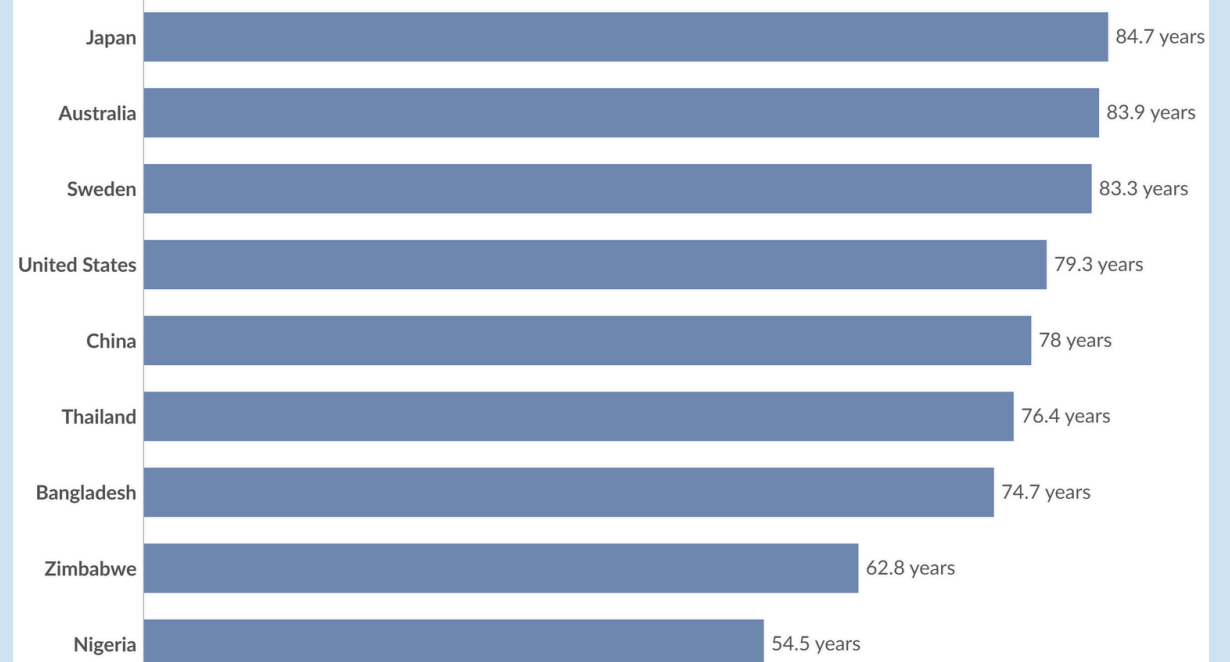
$E[Y]$ hides a lot of heterogeneity that we can see with $E[Y|X]$!



Life expectancy, 2023

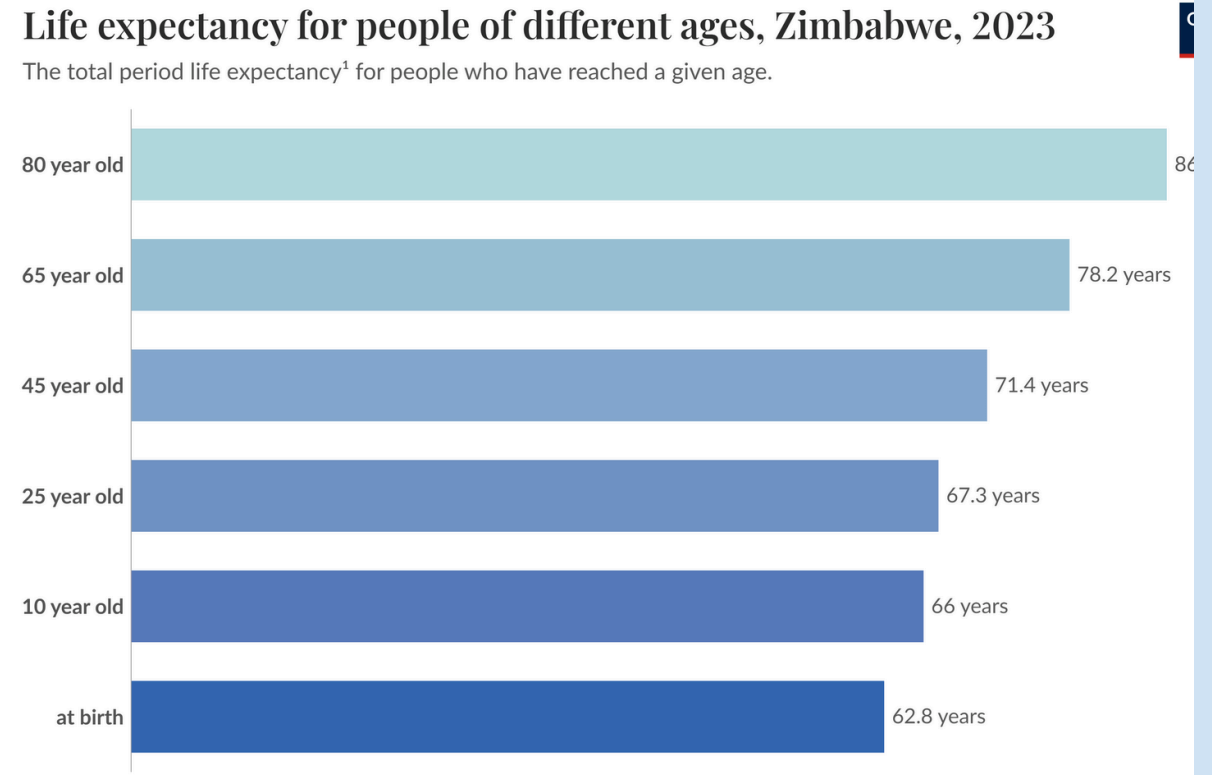
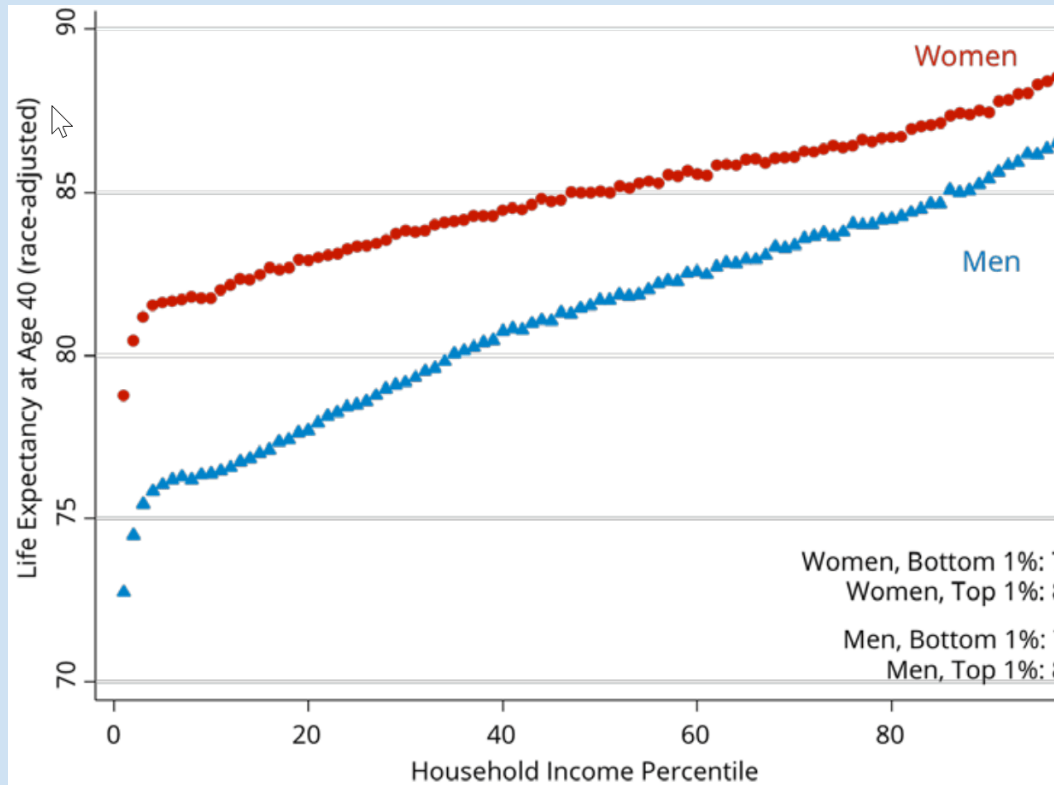
The period life expectancy¹ at birth, in a given year.

Our World
in Data

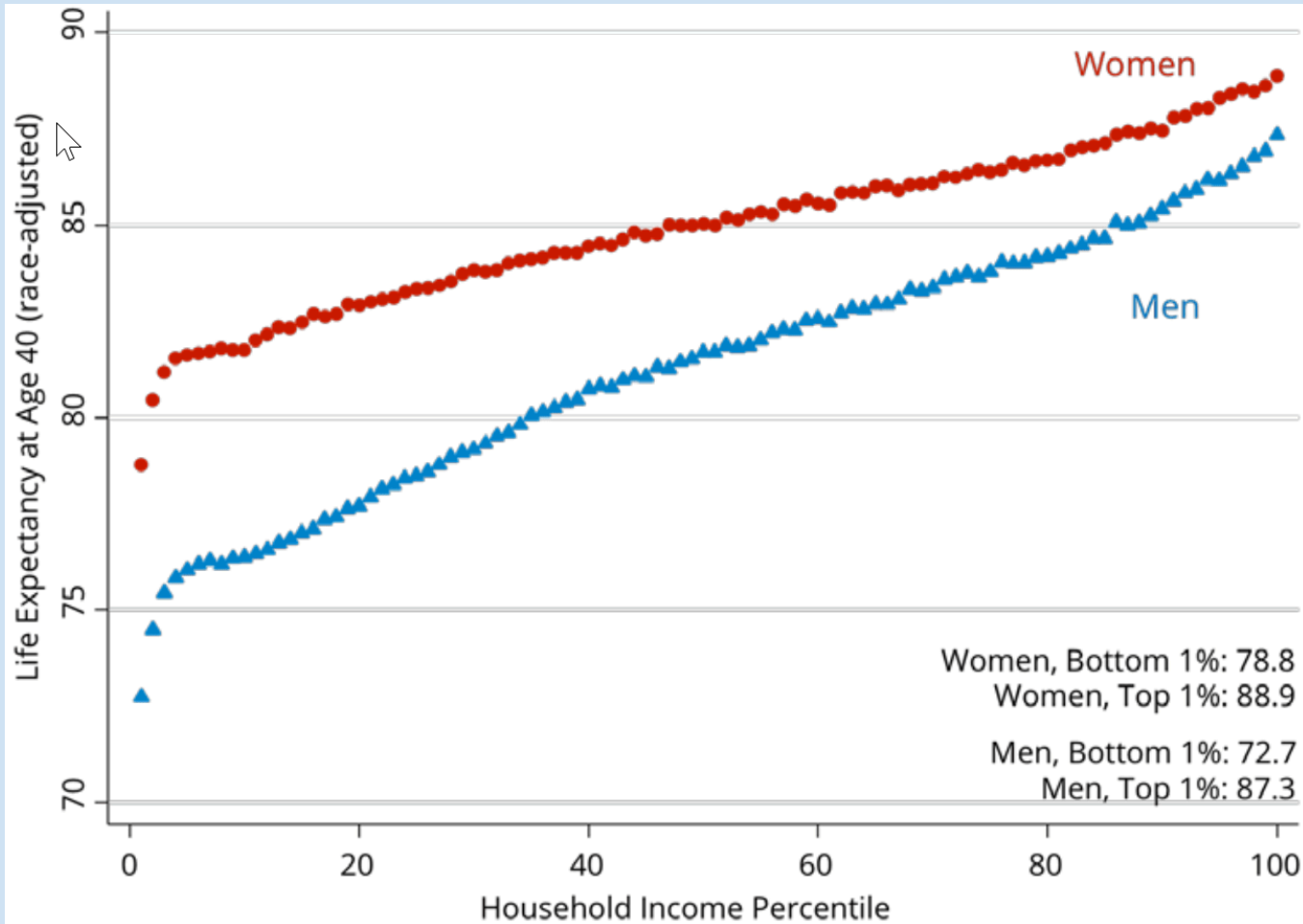


Importance of looking beyond averages!

Low-income men in the US have $\approx$ LE as Zimbabwean men!



Is this a causal relationship?

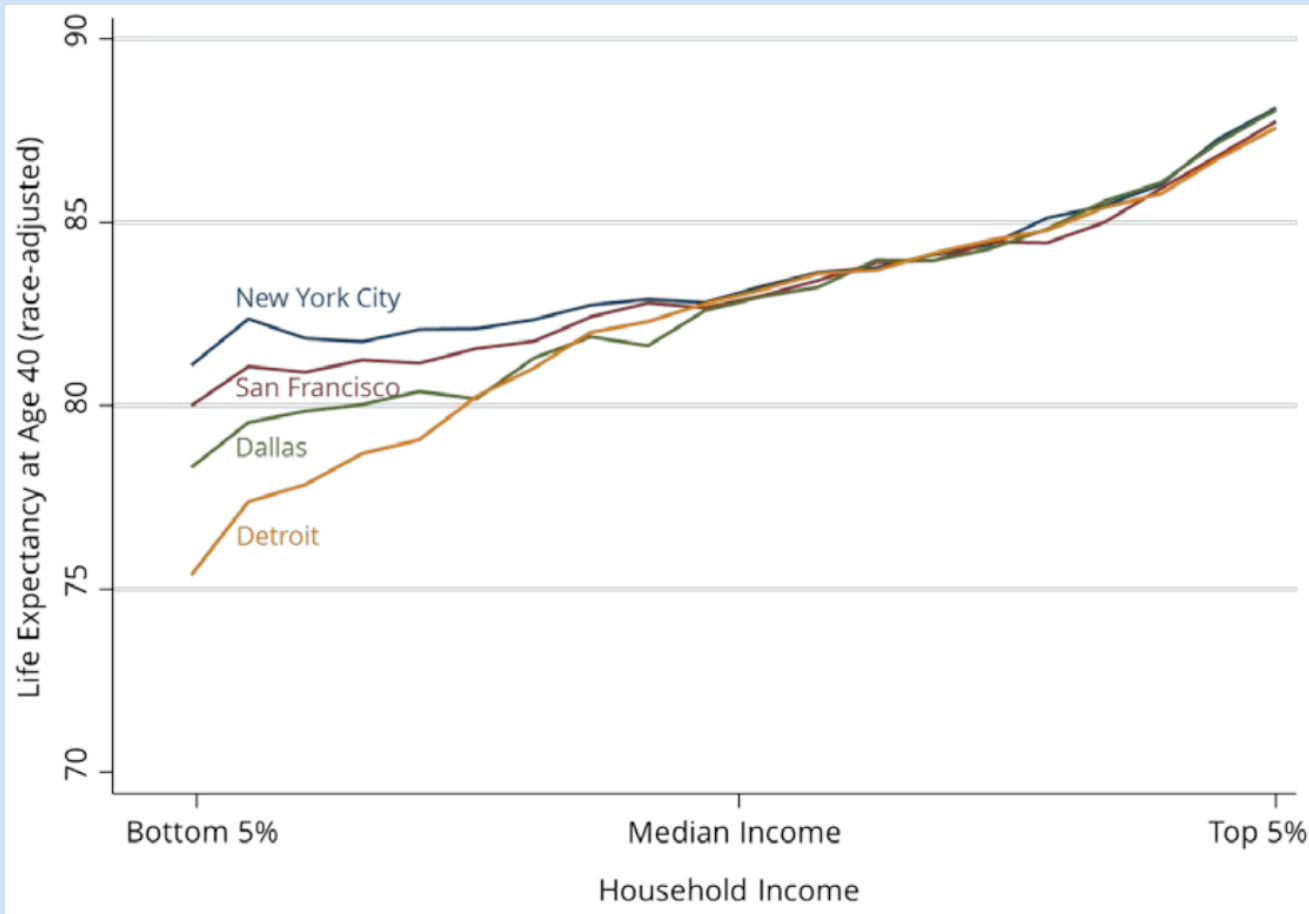


Does this suggest that giving more money to those at the bottom of the income distribution would ↑ their life expectancy?

Source: [Chetty et al. \(2016\)](#)

Descriptives can be incredibly powerful!

Place matters!



It's better to be poor in
New York than in Detroit

Source: [Chetty et al. \(2016\)](#)

Discrete, continuous, or
categorical?

The recording's rule needs a companion for real data

Essential Concepts, week 3: discrete takes a finite set of values, continuous takes a continuum

But every column in every dataset has finitely many values

- PISA [escs](#), the socio-economic index: 62 distinct values in 12,136 rows
- PISA [math](#) score: 3,858 distinct values

Ask what the variable measures in the world, not how many values the column has

Three kinds of variables in the world

In the world	Example	Arithmetic makes sense?
A count	number of siblings: 0, 1, 2, ...	yes: an average of 1.4 siblings
A measurement	math score, socio-economic index, height	yes: any value in a range, rounded when recorded
A label	degree enrolled in, suburb, gender	no: there is no average suburb

Codes look like numbers, and Python will happily average them

Education recorded as 1 = completed primary, 2 = completed high school, 3 = completed a degree

`df['education'].mean()` returns 2.3

2.3 is a fact about the codes, not about anyone's education

Rounding keeps the arithmetic, binning destroys it

Rounded

- A score of 61.4 recorded as 61
- Average of 61, 74, 58: 64.3, and it means something

Binned

- PISA [book](#): books at home, recorded as 0-10, 11-25, 26-100, 101-200, 201-500, 500+
- Average of "101-200" and "26-100": nothing

Books at home is a count in the world; the survey turned it into a label at collection time

Two words for the label case, both in use

- The Essential Concepts recordings say discrete for anything that is not continuous
- Categorical is the everyday word for the label kind: it names why averaging is nonsense
- You will meet both; when a variable is a label, say categorical and nobody will object

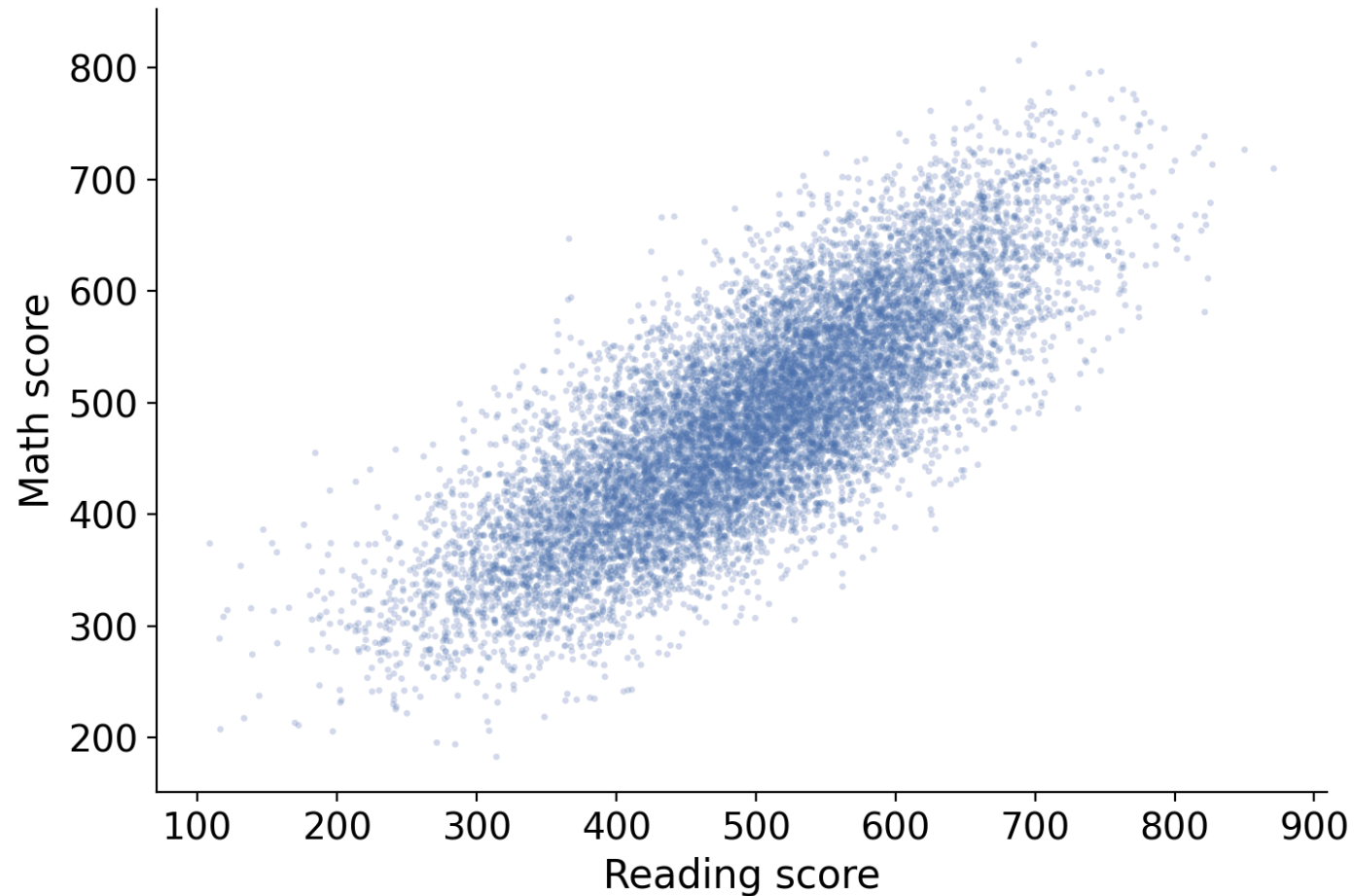
The correlation gallery

Six clouds of dots

For each one: tell your neighbor your guess, then slide to your own answer on the portal

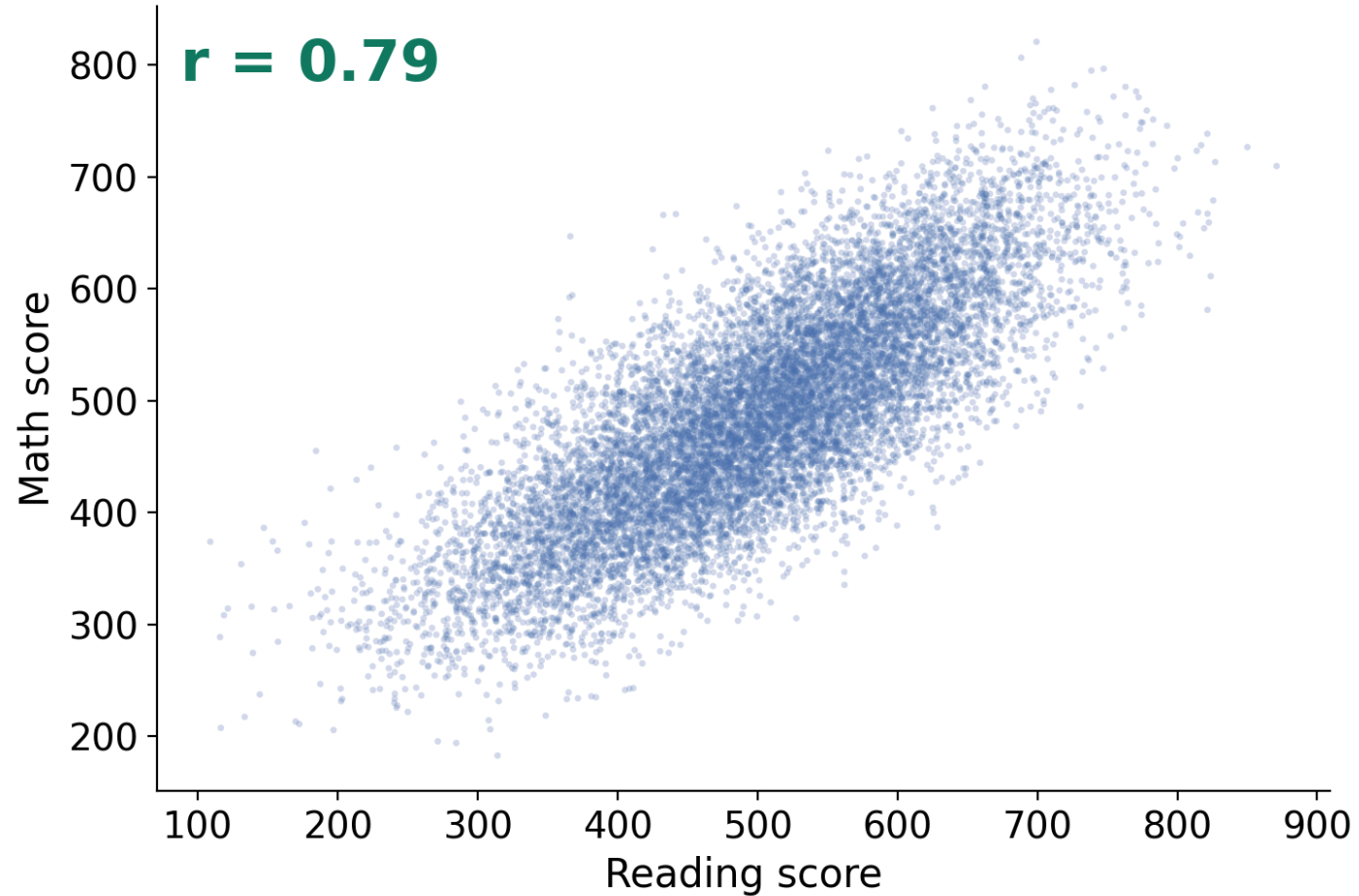
Then we see the room's guesses against the truth

Gallery 1: math and reading, 12,136 Australian students

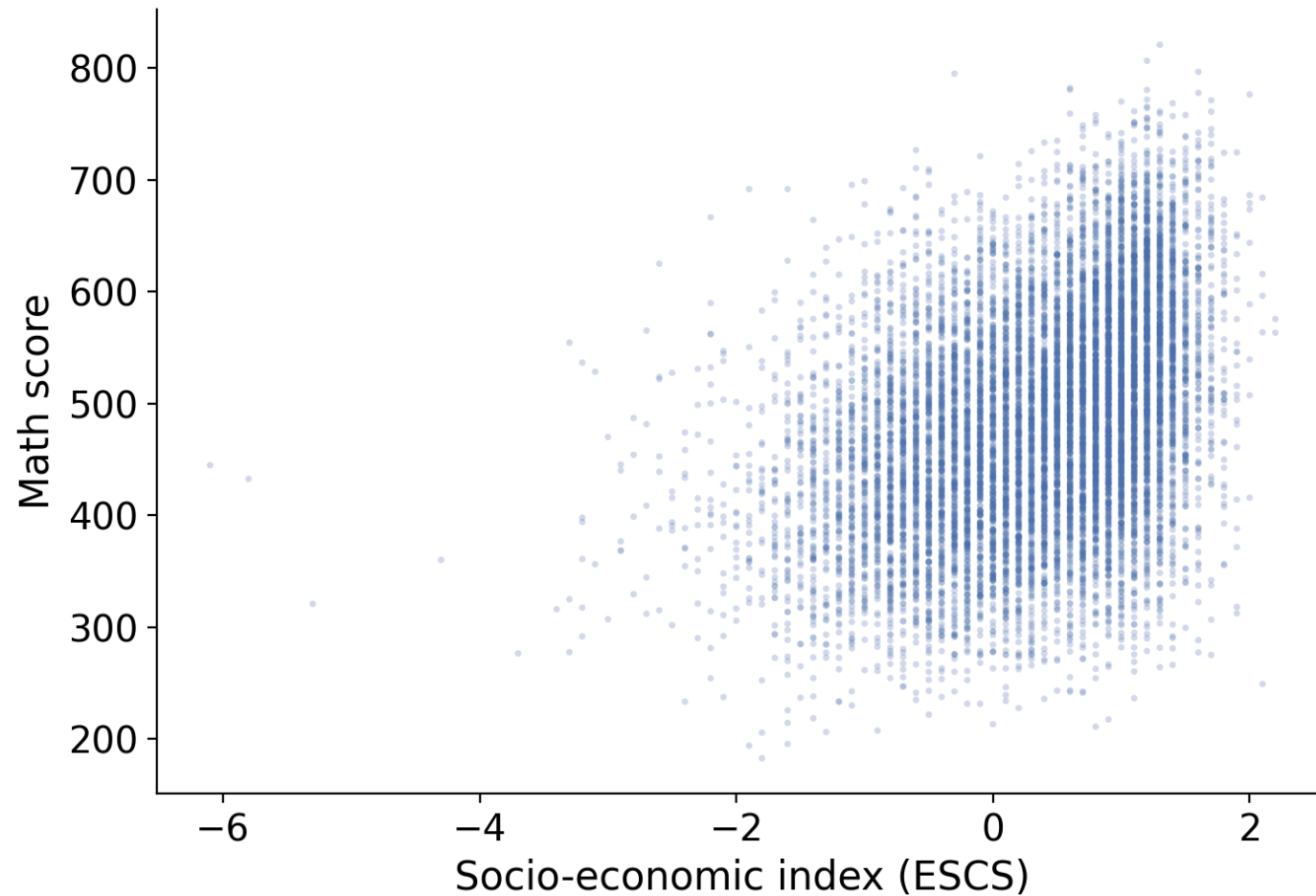


Data: PISA 2022, Australia (OECD)

Gallery 1: math and reading, $r = 0.79$

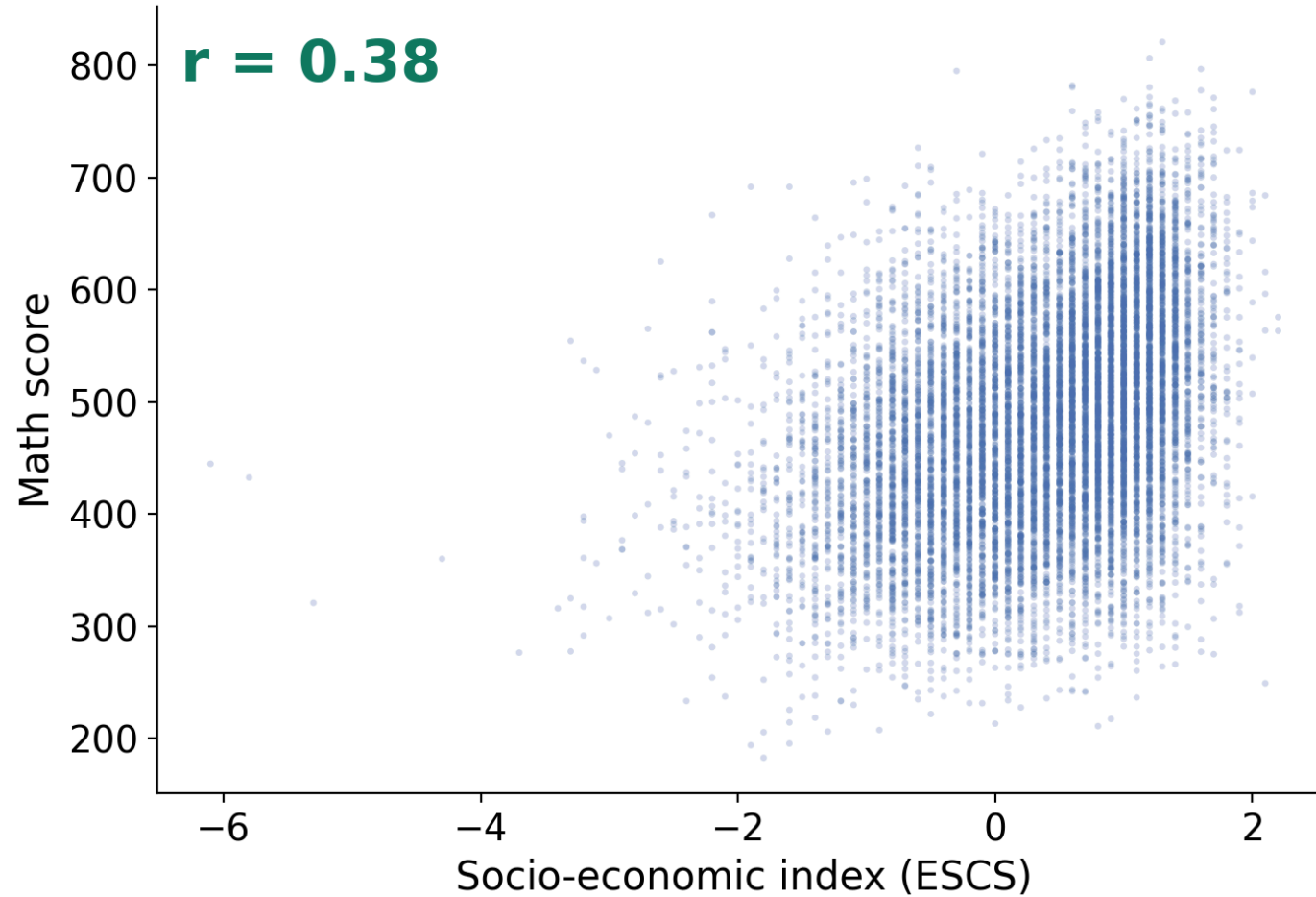


Gallery 2: math and family background, same students

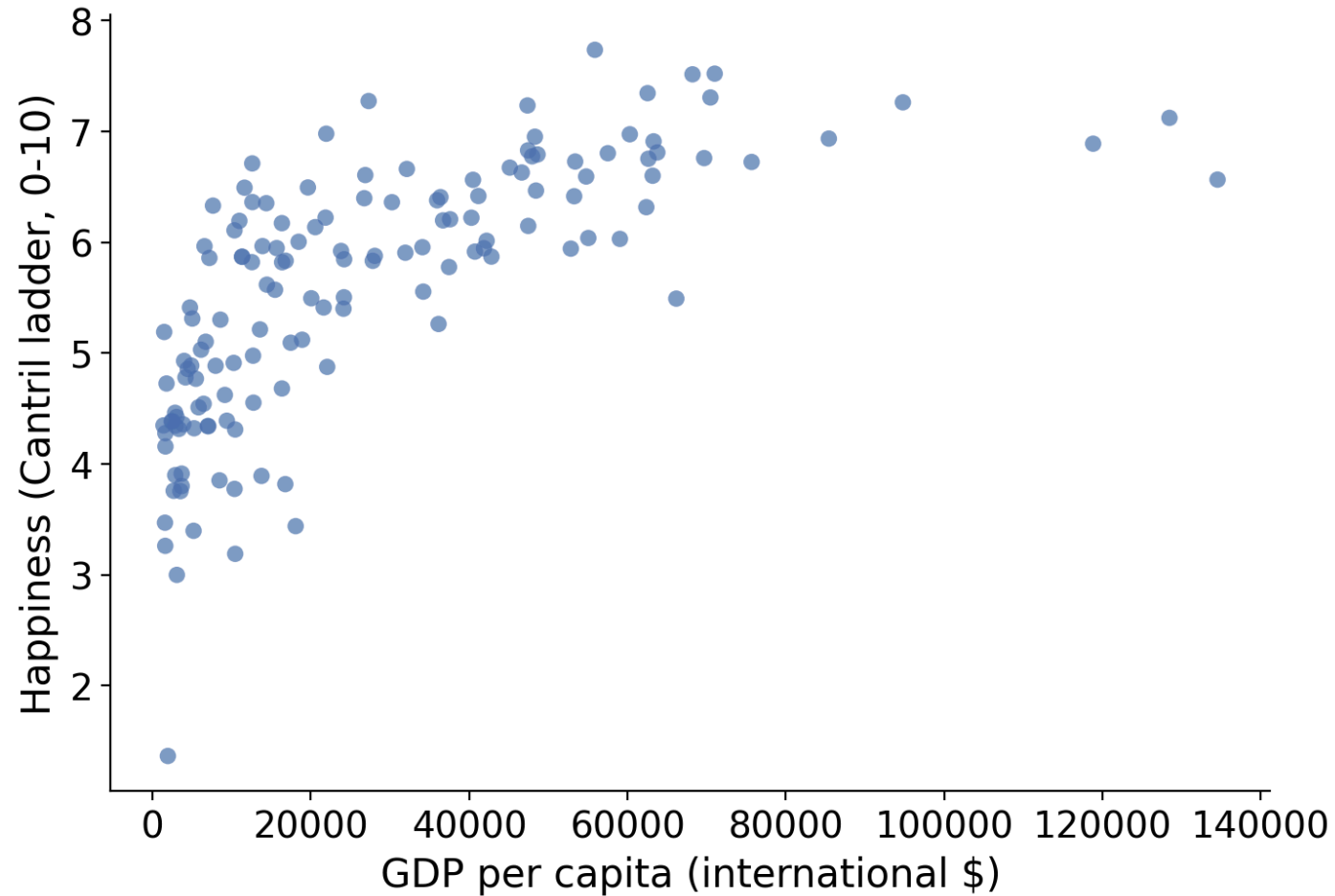


Data: PISA 2022, Australia (OECD)

Gallery 2: math and family background, $r = 0.38$

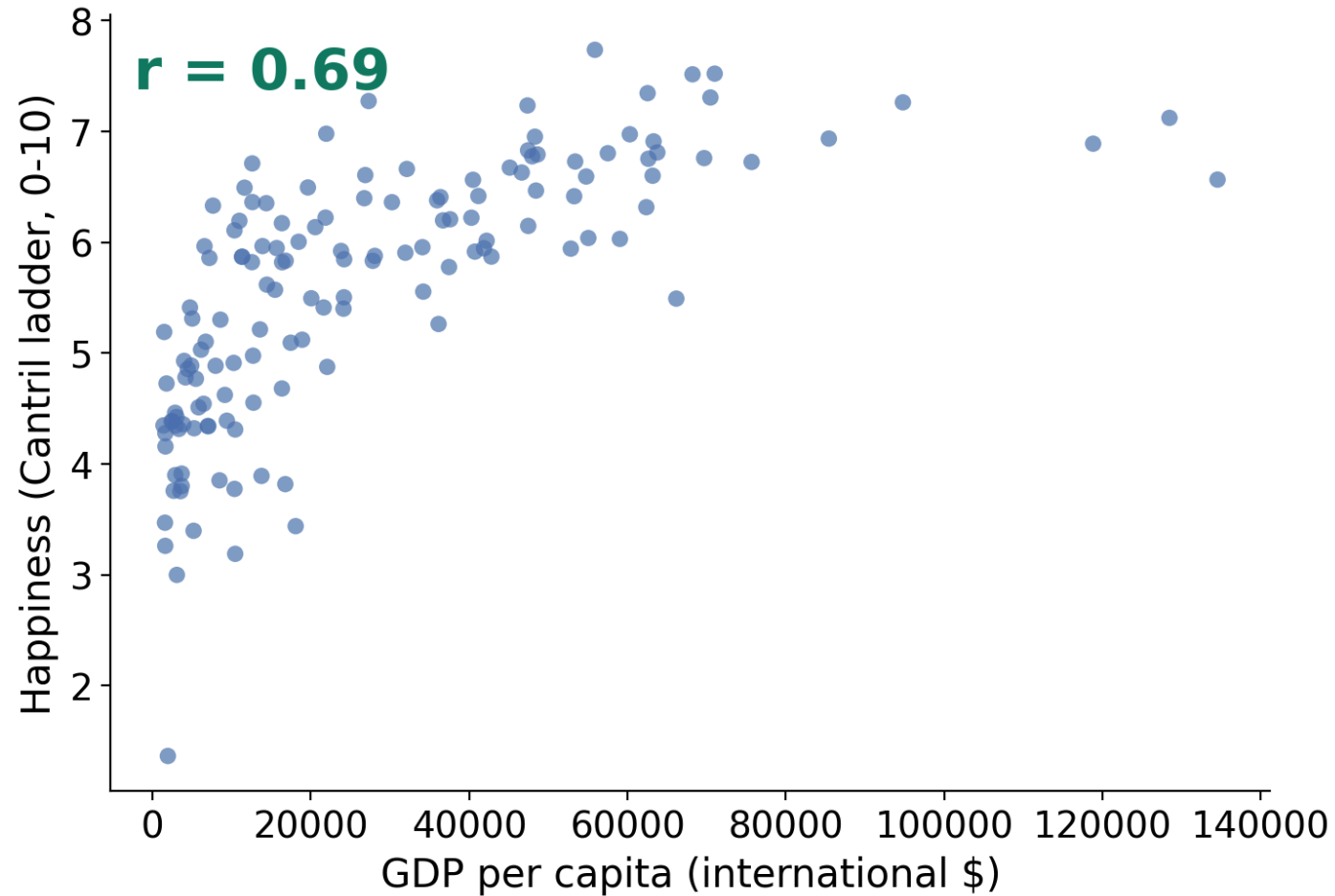


Gallery 3: happiness and GDP per capita, 144 countries



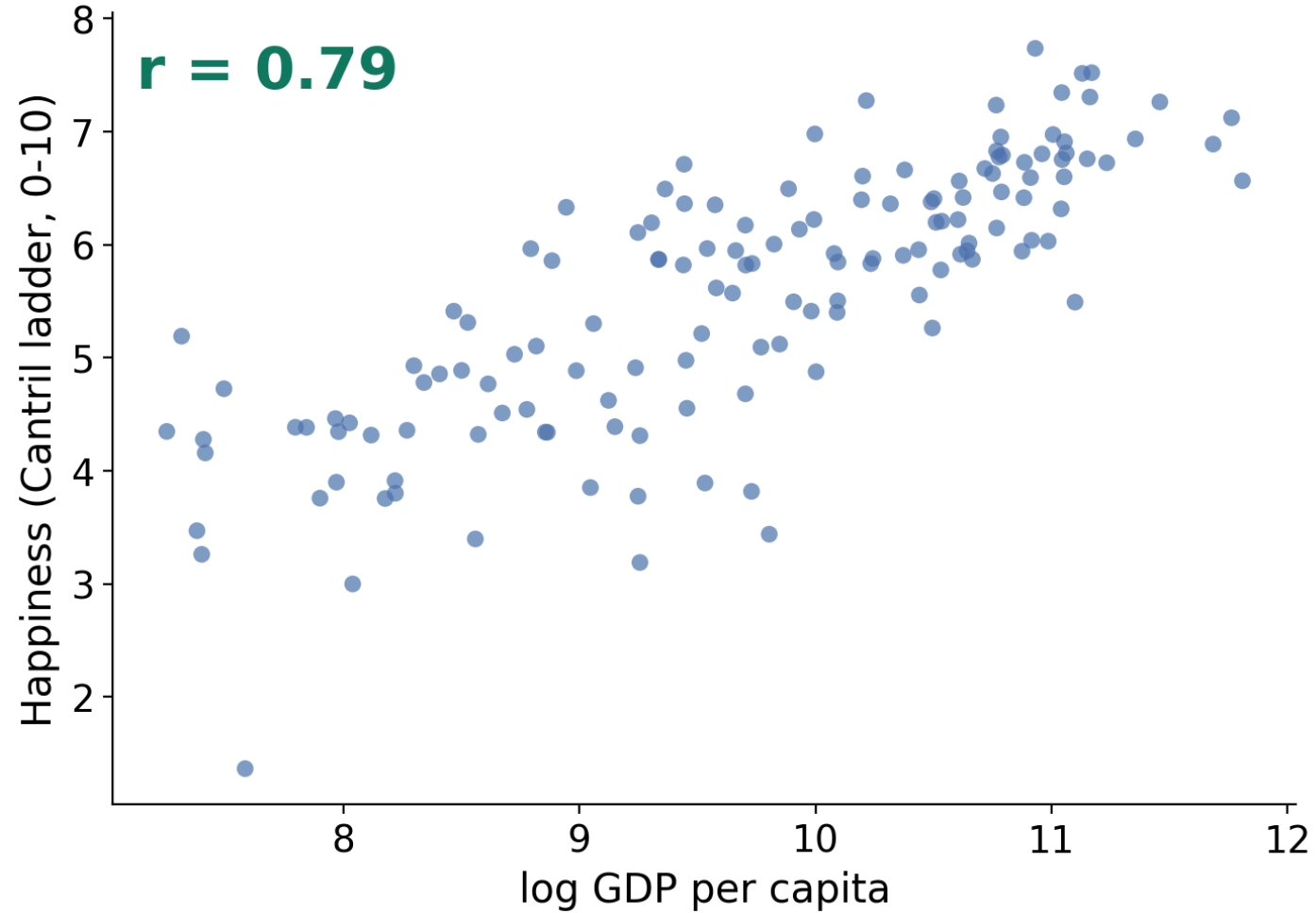
Data: Gallup World Poll & World Bank, via Our World in Data

Gallery 3: happiness and GDP per capita, $r = 0.69$

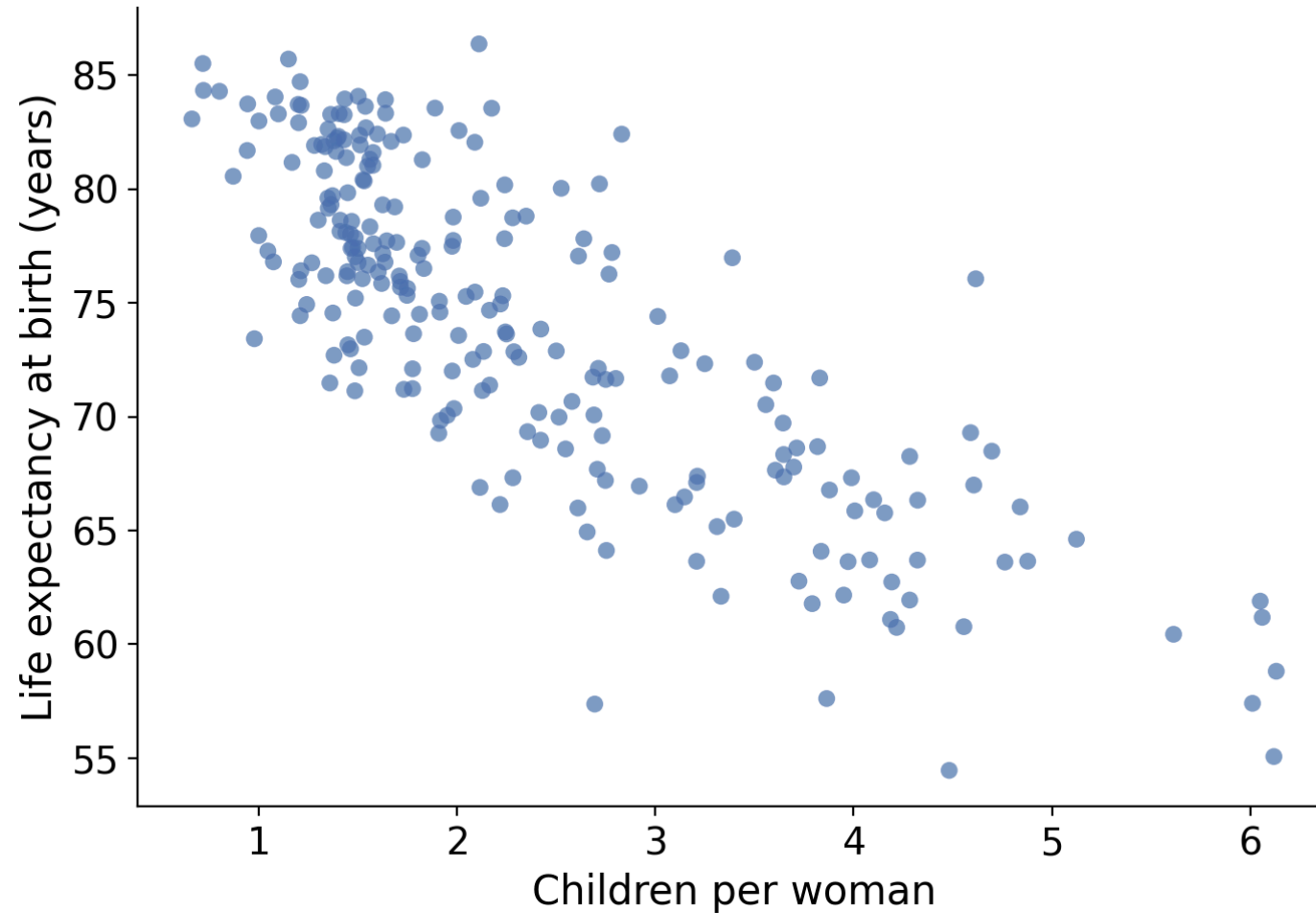


Data: Gallup World Poll & World Bank, via Our World in Data

Same countries, log GDP: $r = 0.79$

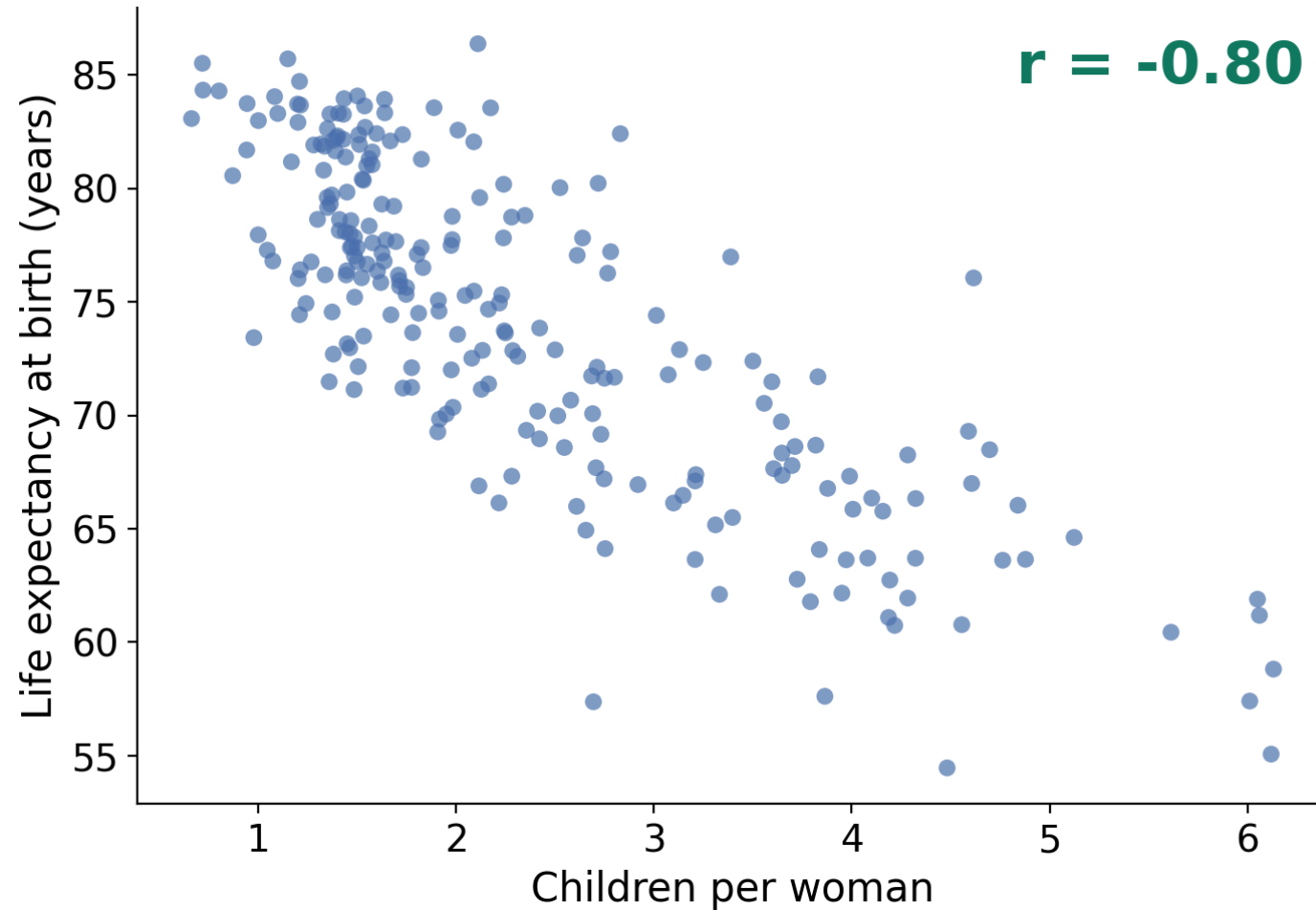


Gallery 4: fertility and life expectancy, 236 countries

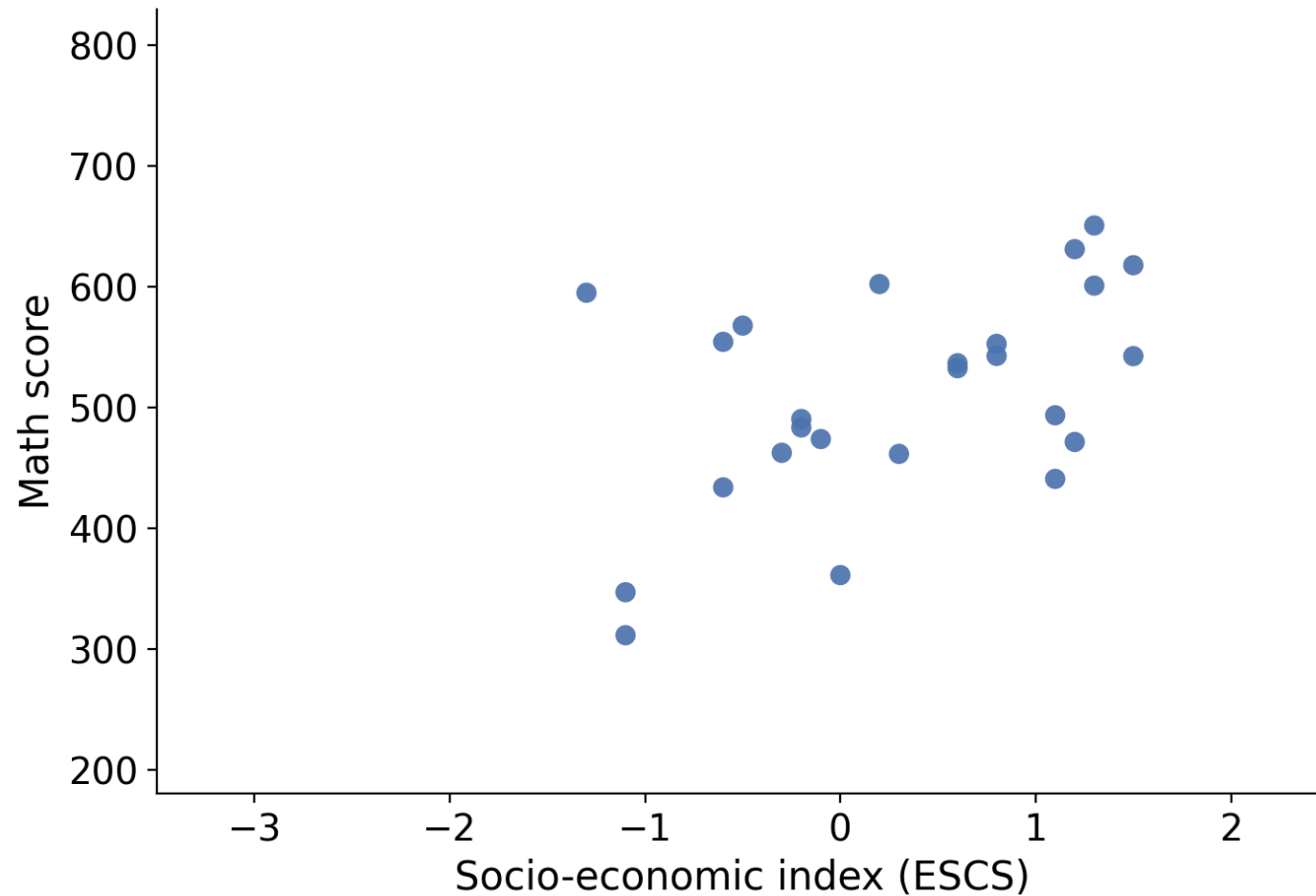


Data: UN World Population Prospects, via Our World in Data

Gallery 4: fertility and life expectancy, $r = -0.80$

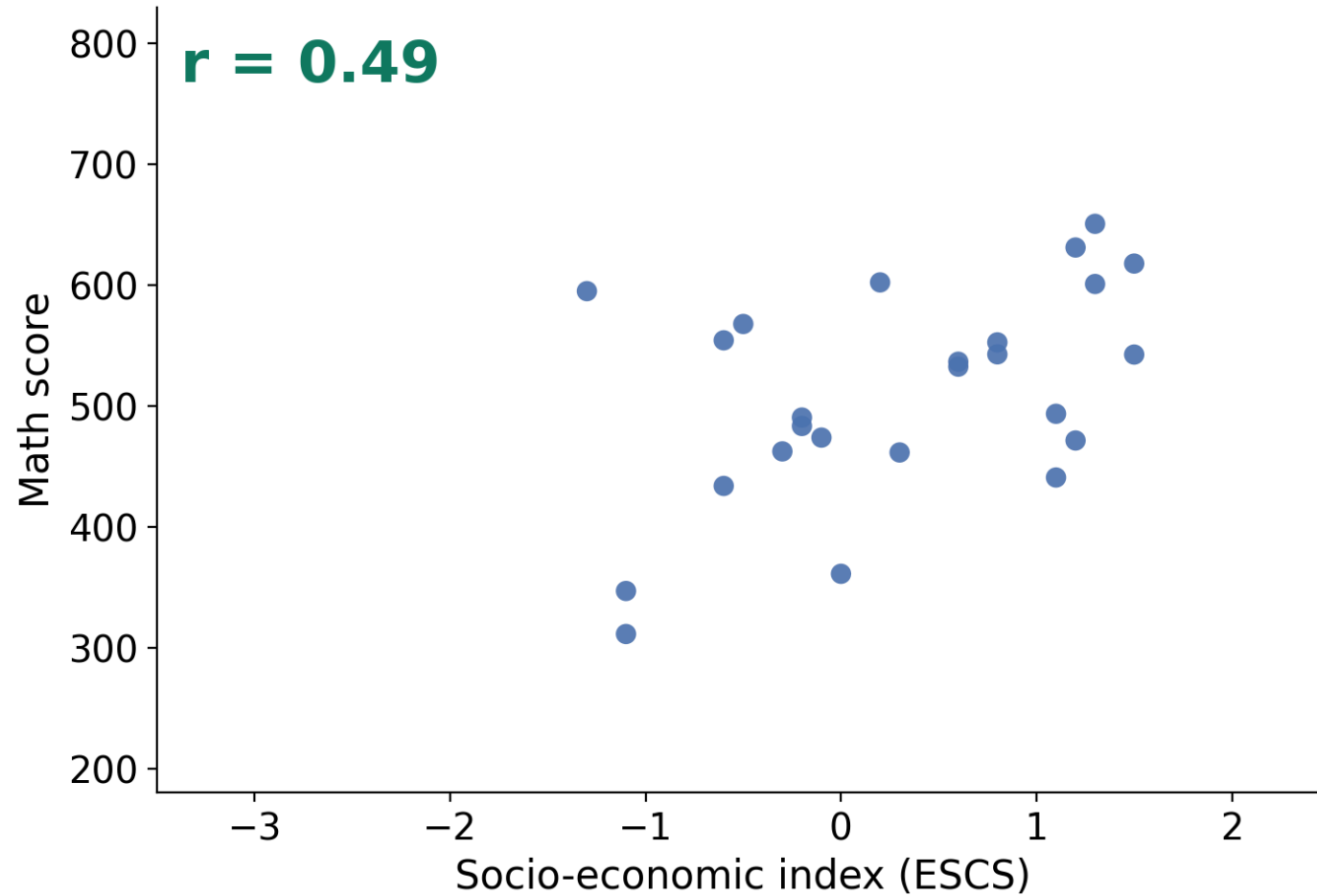


Gallery 5: math and family background again, 25 students



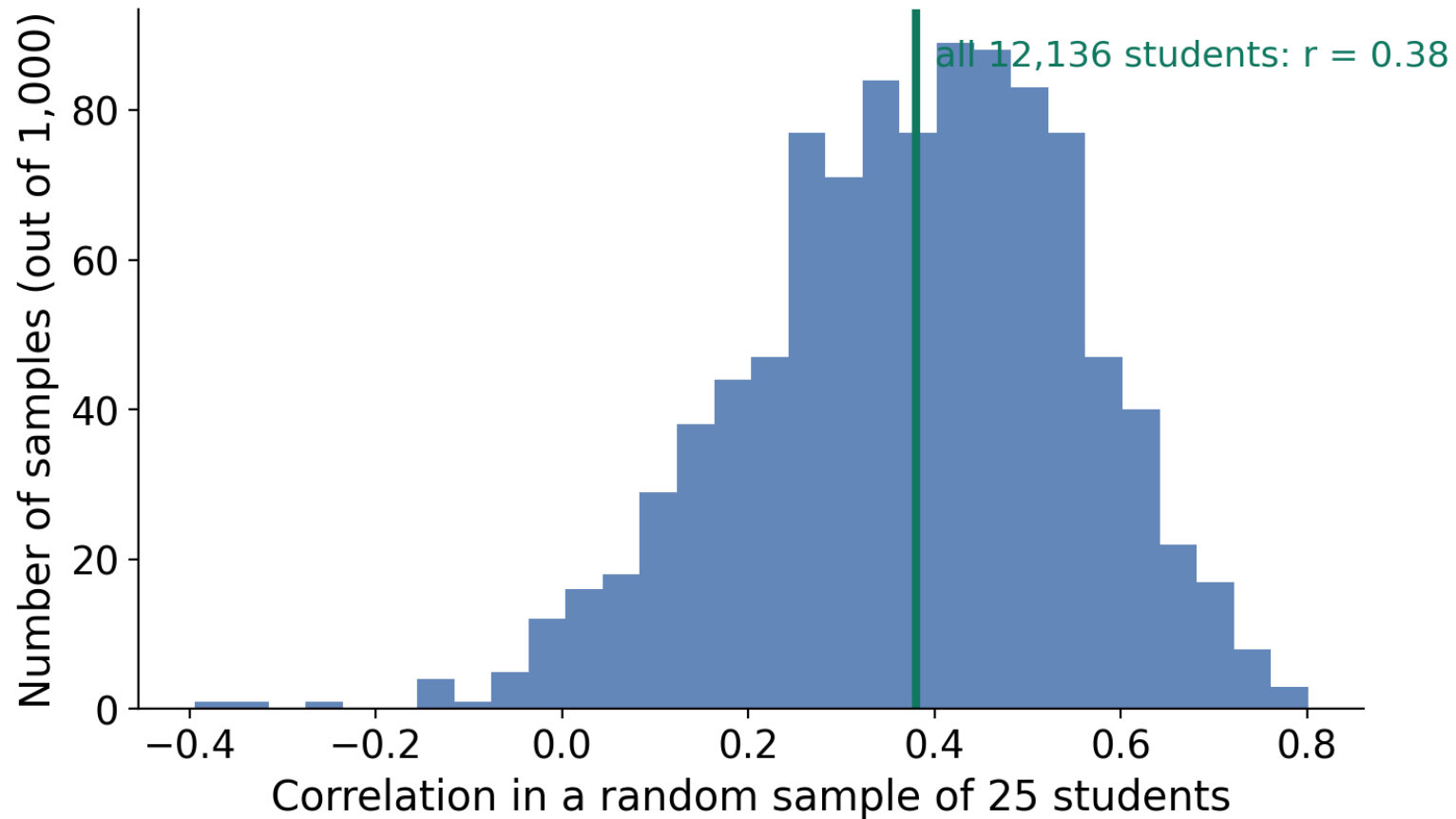
Data: PISA 2022, Australia (OECD), 25 students drawn at random

Gallery 5: 25 students, $r = 0.49$

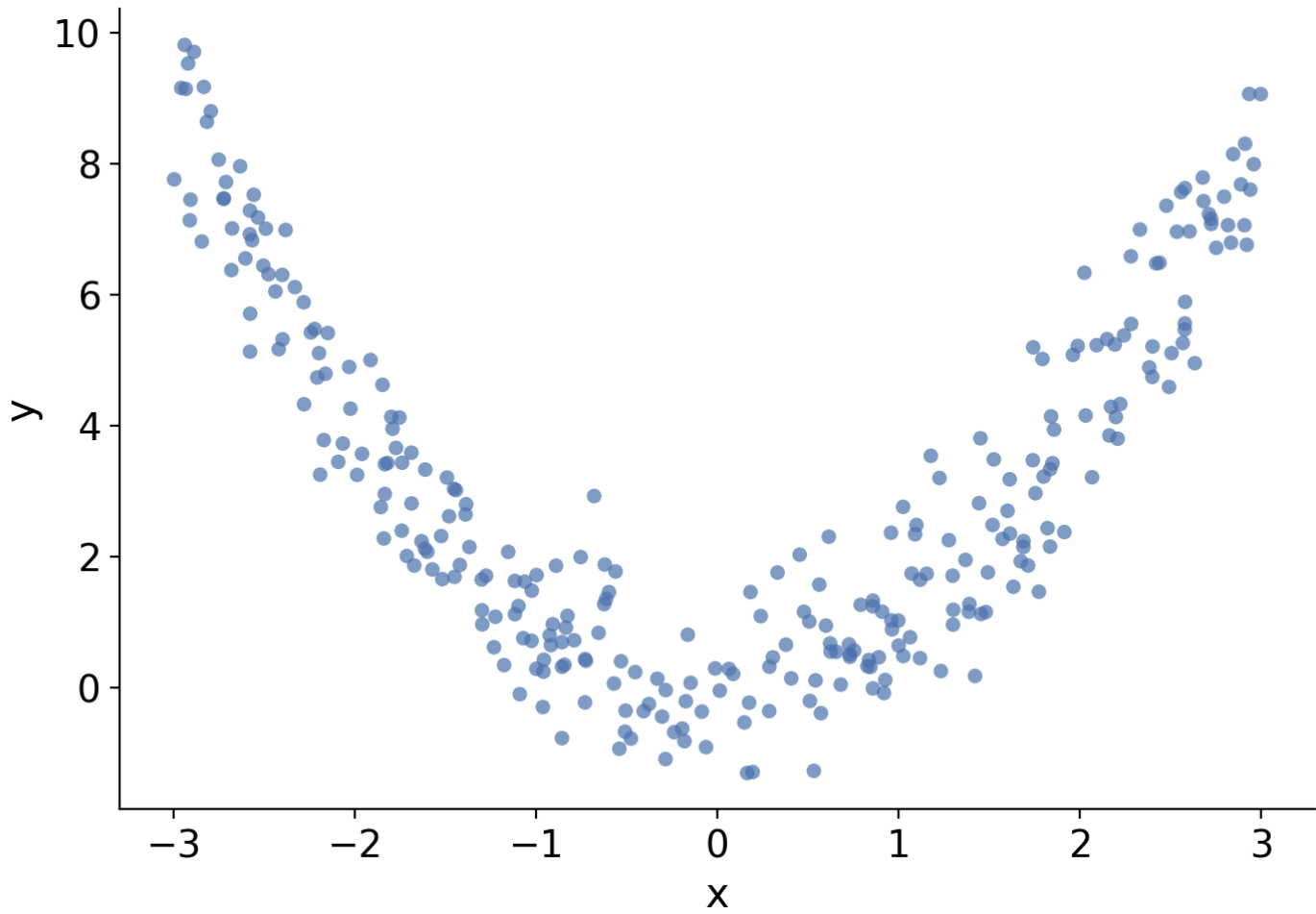


Data: PISA 2022, Australia (OECD), 25 students drawn at random

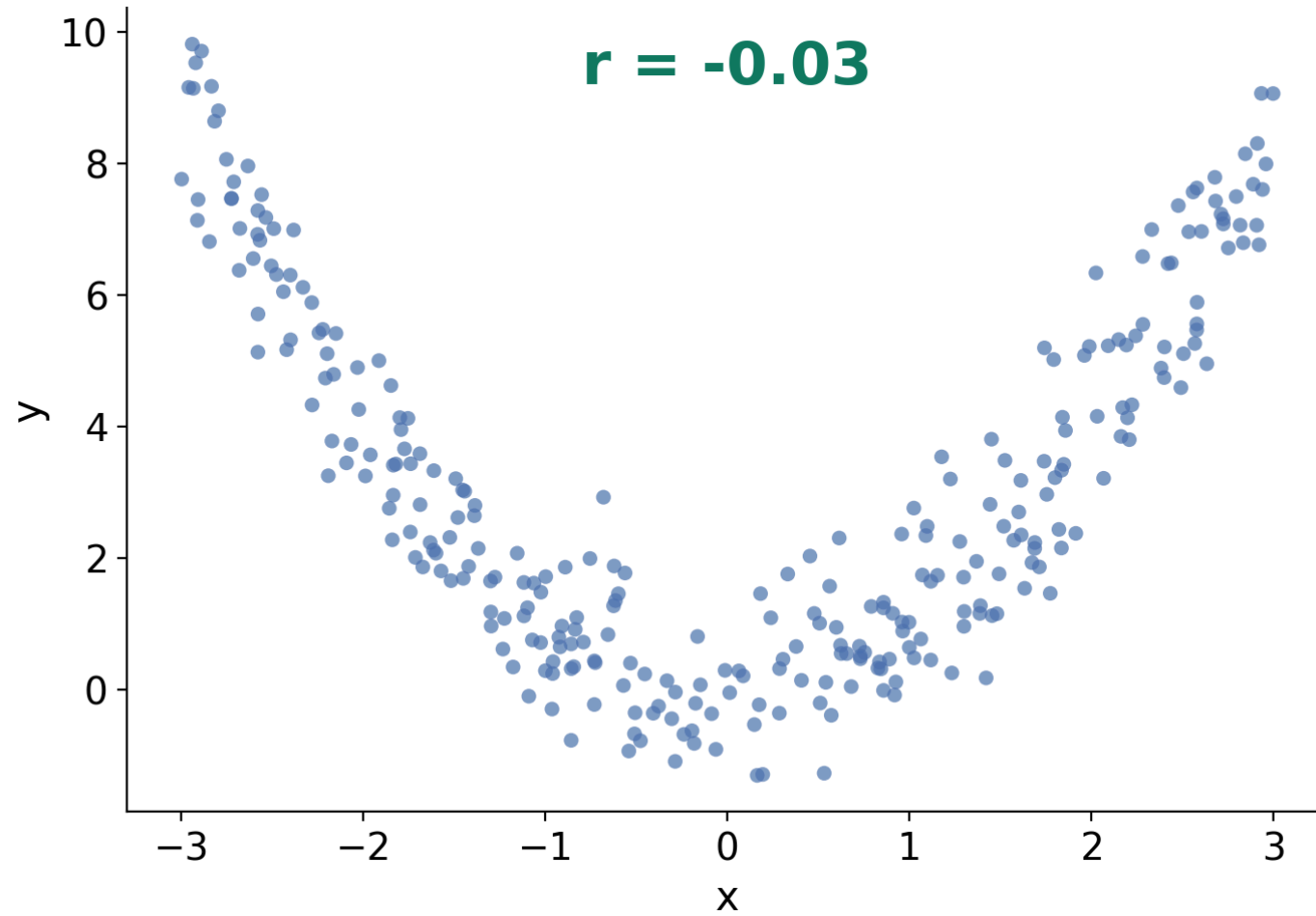
A thousand samples of 25 students: r wanders from -0.4 to 0.8



Gallery 6: one last cloud

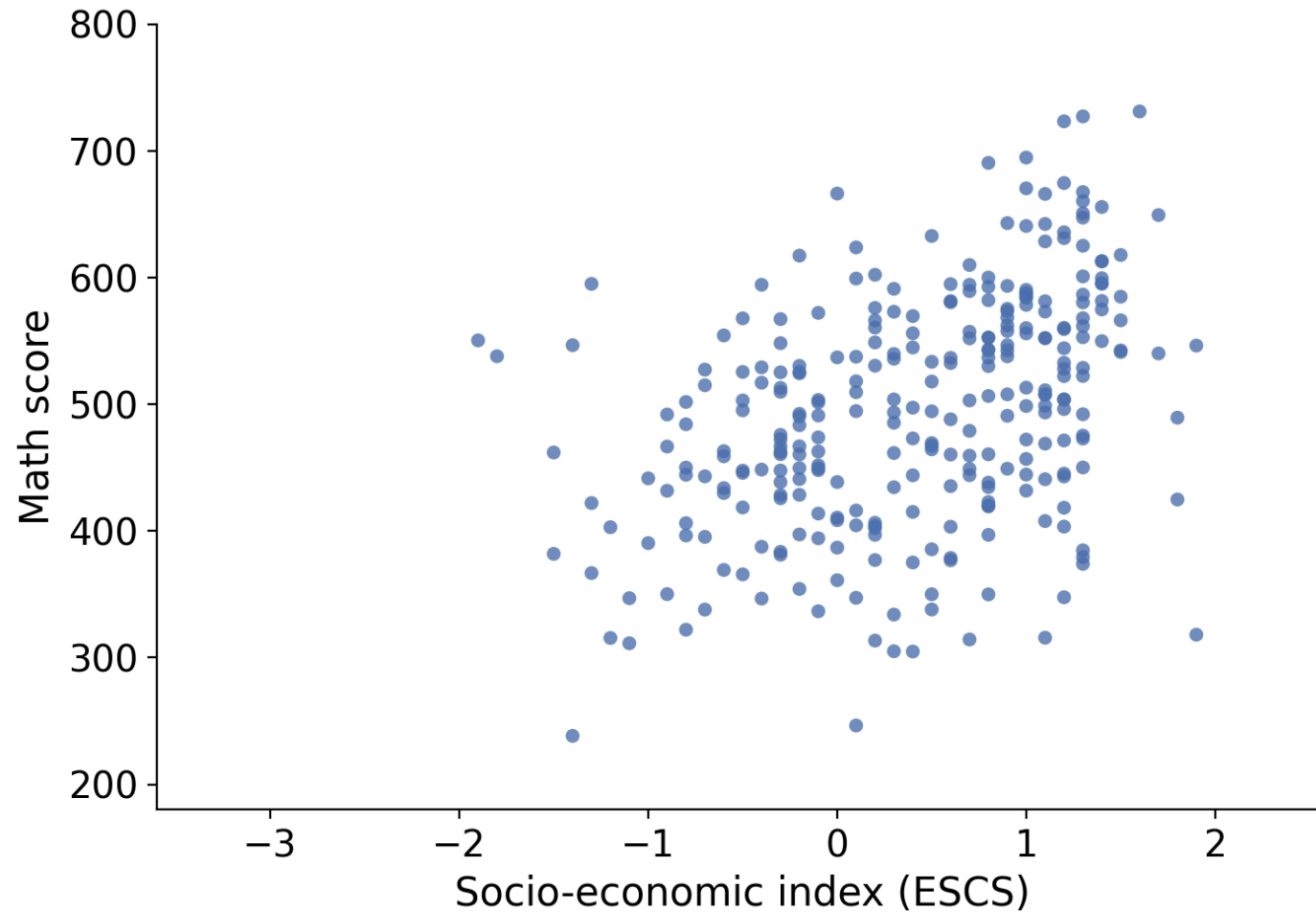


Gallery 6: $r = -0.03$, yet y is a function of x



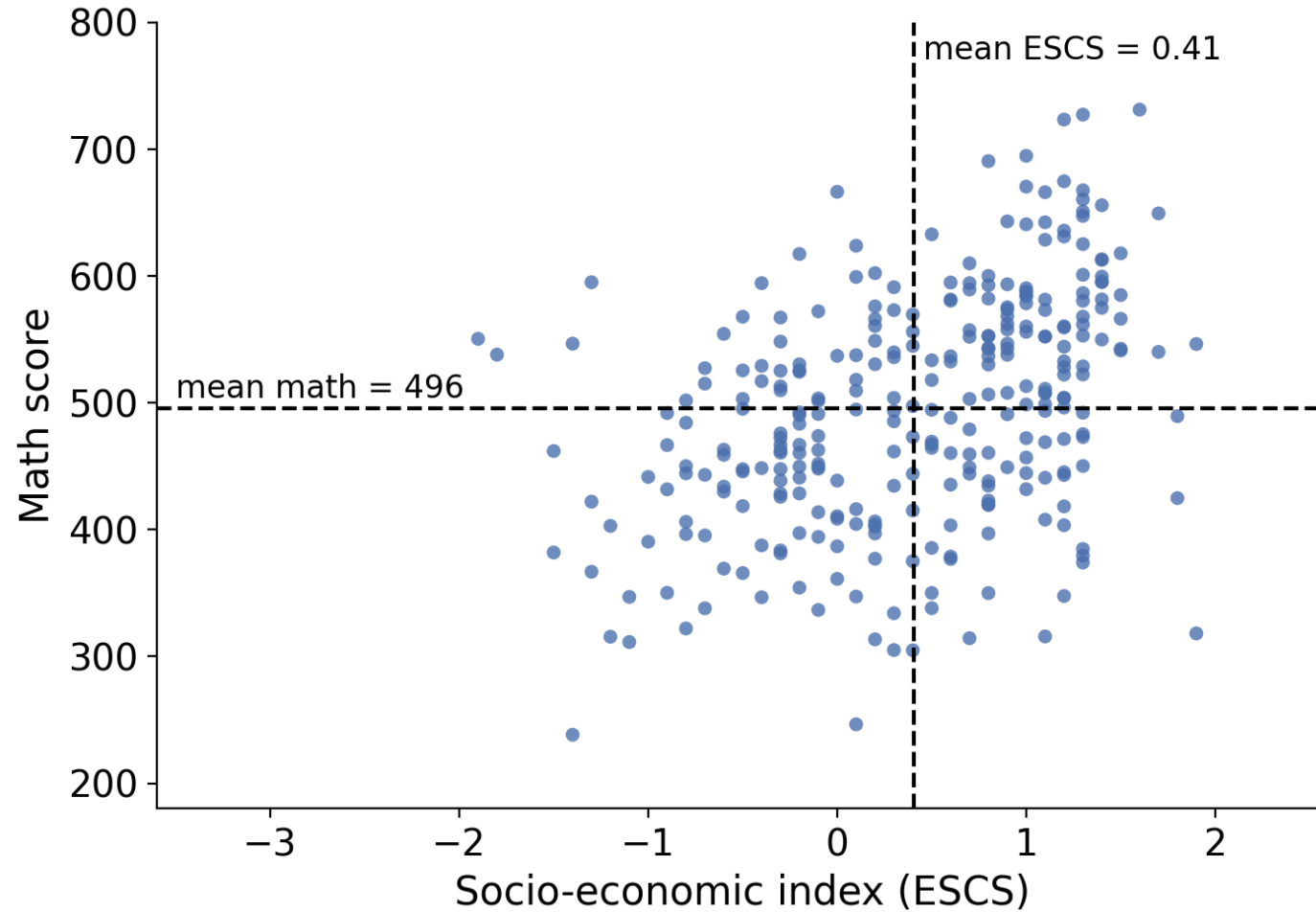
Where a covariance comes from

Three hundred students, math score against family background

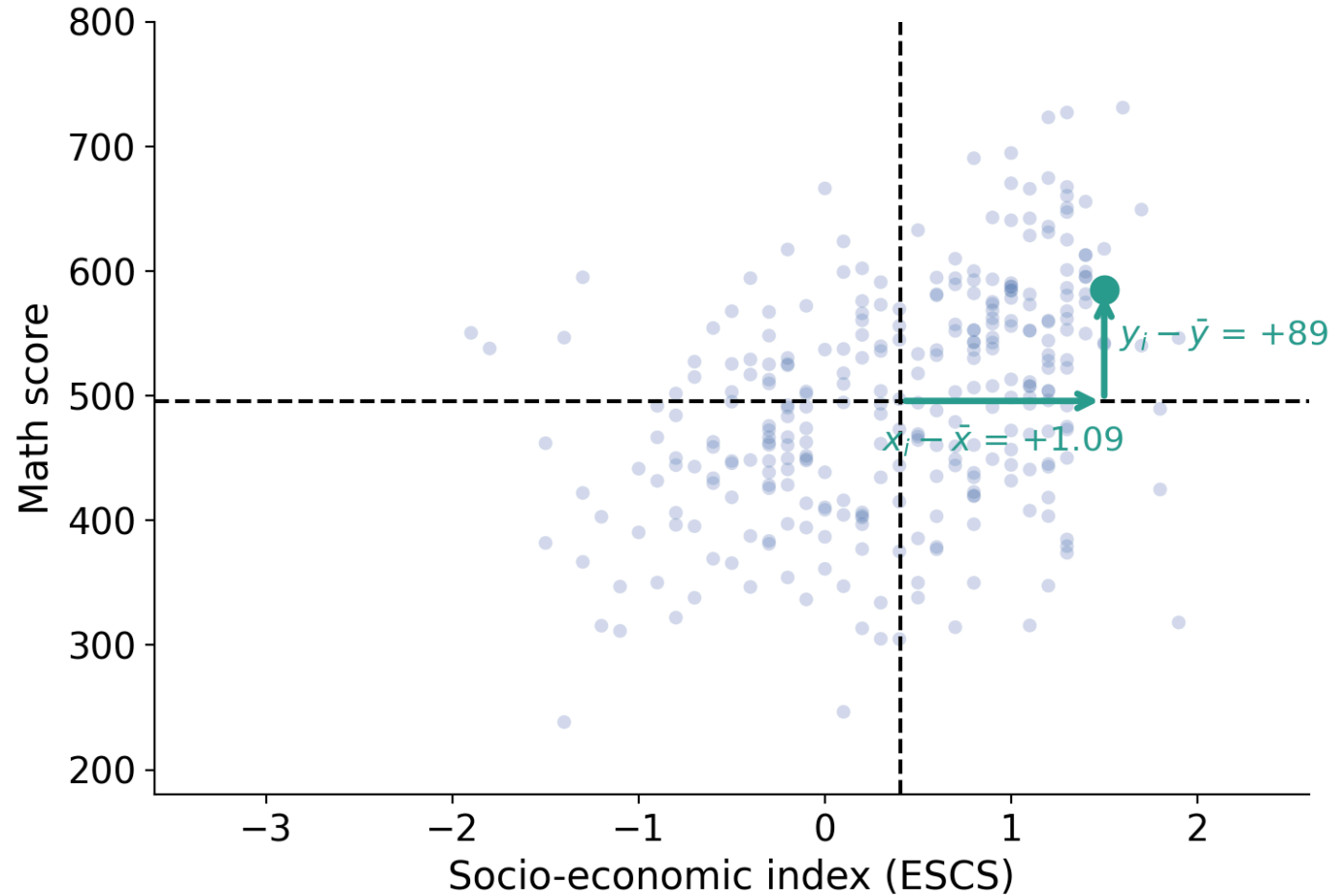


Data: PISA 2022, Australia (OECD), 300 students drawn at random

Step 1: find the two means

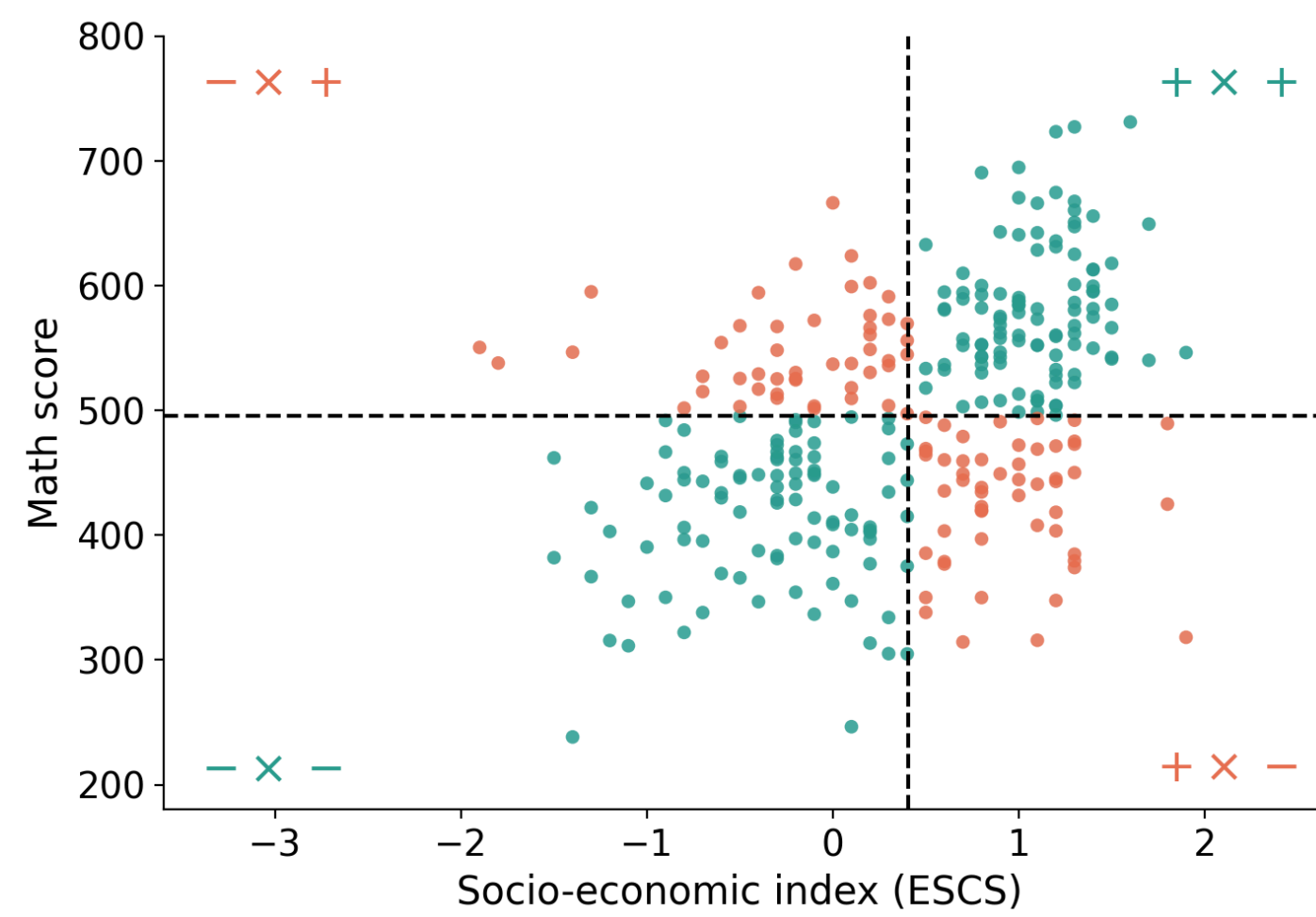


Step 2: each student's two deviations

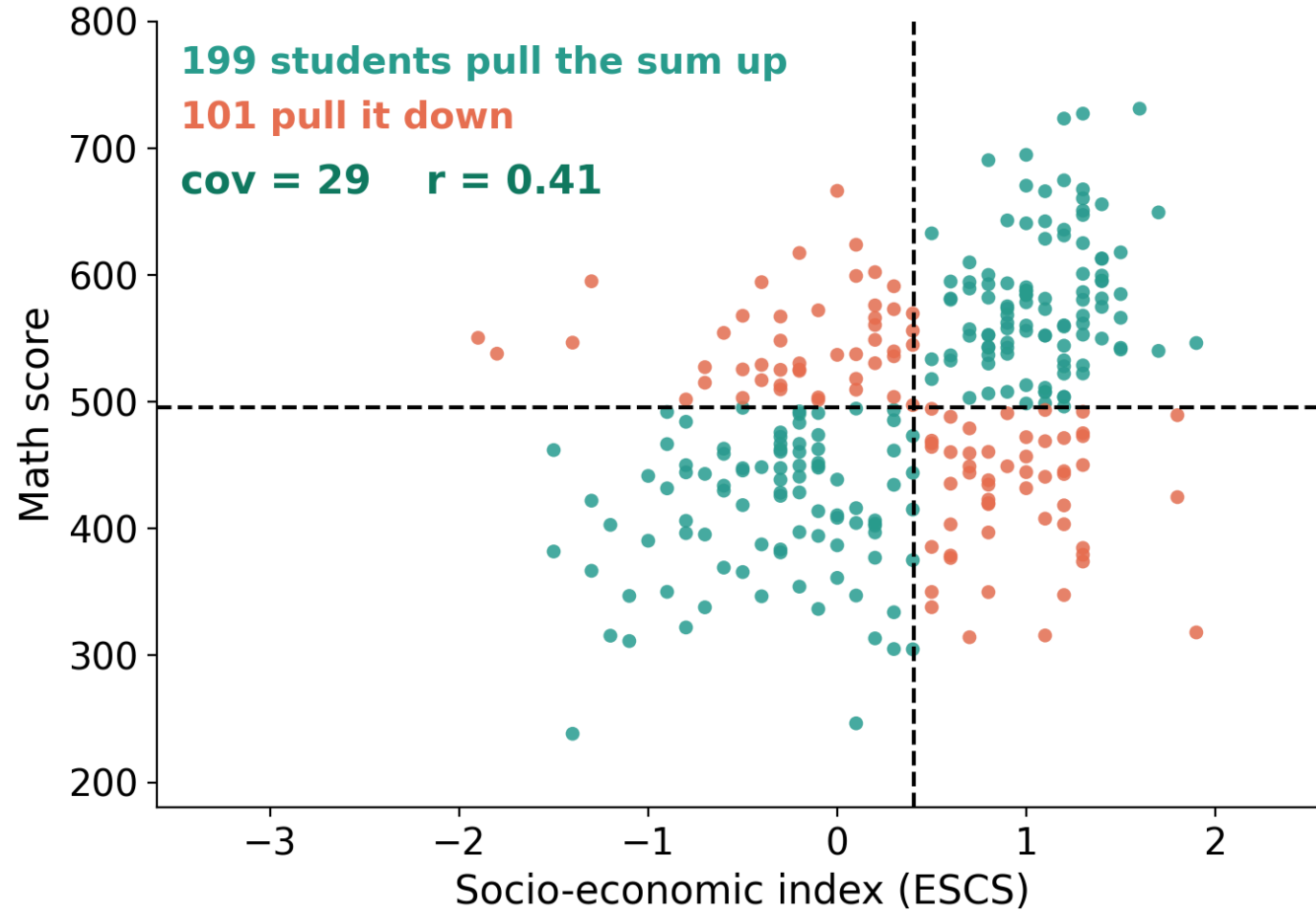


$$(x_i - \bar{x})(y_i - \bar{y}) = (+1.09)(+89) = +97$$

Step 3: the sign of the product is the quadrant



Step 4: add them up: 199 pull up, 101 pull down



The covariance is in nonsense units, so we standardize

$$\text{cov}(X, Y) = \frac{1}{n} \sum_i (x_i - \bar{x})(y_i - \bar{y}) = 29 \text{ index points} \times \text{score points}$$

$$r = \frac{\text{cov}(X, Y)}{s_X s_Y} = \frac{29}{0.78 \times 91.7} = 0.41$$

Some take-aways

- A conditional expectation is a function: one average for every value of X
- Averaging hides the ends: 14.6 years, inside one country
- Correlation sees straight lines only, and a sample's r is itself random
- The covariance is a vote by quadrant; dividing by the SDs removes the units